

**California Regional Water Quality Control Board
Santa Ana Region**

**Order No. R8-2018-0070
Amendment of Order No. R8-2015-0024, NPDES No. CA8000326
Waste Discharge Requirements and Master Reclamation Permit
For
Irvine Ranch Water District
Water Recycling Plants
Surface Water Discharges and Recycled Water Use
Orange County**

The California Regional Water Quality Control Board, Santa Ana Region (hereinafter, Santa Ana Regional Water Board), finds that:

1. On June 19, 2015, the Santa Ana Regional Water Board adopted Order No. R8-2015-0024, NPDES No. CA8000326, prescribing waste discharge requirements and master reclamation permit for the Irvine Ranch Water District's (hereinafter Discharger or IRWD's) water recycling plants, surface waters discharges and recycled water use within the Irvine Groundwater Management Zone.
2. The Discharger owns and operates the Michelson and Los Alisos Water Recycling Plants (WRPs) and a recycled water distribution system that includes lined and unlined reservoirs, pump stations, and pipelines, among other features, that distributes recycled water within the boundaries of the Santa Ana Regional Water Board and the San Diego Regional Water Quality Control Board (San Diego Regional Water Board). However, IRWD's water recycling plants, reservoirs and most of the recycled water use areas are located within the boundaries of the Santa Ana Regional Water Board and a small fraction of recycled water use areas are located within the San Diego Regional Water Board.
3. Currently, the San Diego Regional Water Board regulates the use of recycled water, produced at the Discharger's WRPs, within the Aliso Creek Hydrologic Sub-area under Order 97-52, *Waste Discharge and Water Recycling Requirements for the Production and Purveyance of Recycled Water by Member Agencies of the South Orange County Reclamation Authority*, which is a regional water recycling permit.
4. In a letter entitled, "Request for Santa Ana Regional Water Quality Control Board to Regulate Waste Discharge Requirements from Irvine Ranch Water District Water Recycling Plants," dated March 30, 2016, the Discharger requested to this Regional Water Board and the San Diego Regional Water Board the designation of the Santa Ana Regional Water Board as the sole Regional Water Board for the regulation of the treatment, distribution and use of recycled water produced at the Discharger's WRPs. The Discharger made the

regional water board designation request pursuant to Section 13228(a) of the California Water Code, which states the following:

(a) Concerning any matter that may be submitted to a regional board by a person or entity that is subject to regulation by more than one Regional Board, the person or entity may submit the matter to one of those regional boards if both of the following requirements are met:

(1) The person or entity submits a written request to all affected regional boards that one Regional Board be designated to regulate the matter.

(2) All affected Regional Boards agree in writing to the designation. Unless the Board of any affected Regional Board denies the request, the Executive Officer of a Regional Board may grant a request submitted pursuant to paragraph (1) on behalf of that Board.

5. Also, the Discharger's March 30, 2016 letter included supporting information such as Exhibits "A" and "B". Exhibit "A" consists of a map that shows the location of IRWD's water recycling plants, recycled water reservoirs and distribution pipelines within the boundaries of the San Diego and Santa Ana Regional Water Boards. Exhibit "B" contains a comparison between the annual averages of recycled water monitoring data for the last ten years for both IRWD's water recycling plants and the 12-month average effluent limits for recycled water discharges specified under San Diego Regional Water Board's Order 97-52. The recycled water monitoring data indicates that IRWD has complied with all effluent limits for the last ten years.
6. The San Diego and the Santa Ana Regional Water Boards agreed with IRWD's request to designate the Santa Ana Regional Water Board as the sole Regional Water Board for regulating the IRWD's Water Recycling Plants. In a letter dated July 22, 2016, which was addressed to IRWD, the San Diego Regional Water Board memorialized this agreement and included pertinent background and supporting information and conditions for the designation responsibility of the Santa Ana Regional Water Board, which becomes Attachment "P" of Order No R8-2015-0024. The Santa Ana Regional Water Board ratified this agreement in a letter issued to IRWD dated September 19, 2016, which becomes Attachment "Q" of Order No. R8-2015-0024. This designation will become effective once the San Diego Regional Water Board amends Order 97-52 or adopts a new master reclamation permit for the South Orange County Wastewater Authority that excludes IRWD.
7. On July 18, 2017 the Discharger submitted a Report of Waste Discharge (Form 200) with the purpose of requesting authorization to discharge up to 28 MGD of tertiary treated and disinfected effluent, under emergency conditions, to Reach 1 of San Diego Creek through Discharge Point Storm-007 (Storm-007). Through

June 6, 2018, the Discharger submitted several documents in support of the proposed discharge and in response to information request from this Regional Water Board staff to demonstrate that the proposed discharge does not impact San Diego Creek and Newport Bay and complies with the conditions of the Nutrient Total Maximum Daily Load (TMDL) and other TMDLs applicable to the San Diego Creek/Newport Bay watershed and the Water Quality Control Policy for the Enclosed Bays and Estuaries of California. The proposed discharge to Reach 1 of San Diego Creek would occur under emergency conditions where an extreme rainfall event and/or season causes IRWD to run out of recycled water storage, demand for recycled water to be very low, and may not be feasible to send raw sewage, usually treated at Michelson WRP, to Orange County Sanitation District's (OCSD's) wastewater treatment plants (WWTPs) for treatment and disposal.

8. The nutrient TMDL excludes total nitrogen (TN) loads that are discharged to the San Diego Creek/Newport Bay watershed when flow in San Diego Creek is equal or greater than 50 cubic-feet-per-second (cfs) at Campus Drive, however, it does not exclude total phosphorous (TP) loads discharged under any flow conditions in San Diego Creek. The Water Quality Control Policy for the Enclosed Bays and Estuaries of California, as amended by SWRCB's Resolution No. 95-84 on November 16, 1995, prohibits the discharge of new discharges of municipal wastewaters and industrial process waters (exclusive of cooling water discharges) to enclosed bays and estuaries, other than the San Francisco Bay-Delta system, which are not consistently treated and discharged in a manner that would enhance the quality of receiving waters above that which would occur in the absence of the discharge. To comply with this policy and the nutrient TMDL, the Discharger must offset TN and TP loads (lbs/rainfall year) discharged under emergency conditions from the MWRP to Newport Bay at a minimum ratio 1.5:1 in accordance with an approved TN and TP offset plan. The TN and TP offset program will consist of computing and reporting the TN and TP loads that the Discharger removes from the San Diego Creek and Newport Bay Watershed through IRWD's operations of NTS wetlands, including the San Joaquin Marsh, and urban runoff diversions. IRWD must document the TN and TP offsets accumulated over a 3-rainfall year period that includes the rainfall year (October through September) when the effluent discharge occurred. In addition, the Discharger has committed to minimize the TN and TP loads discharged to Reach 1 of San Diego Creek by implementing enhanced nutrient removal with operational objectives of producing average recycled water total nitrogen and total phosphorus concentrations of 10 mg/L and 0.3 mg/L respectively during diversions. The MWRP will be operated in such a manner that the Discharger will meet these TN and TP effluent concentrations as an emergency discharge facility performance goal.
9. Normally, the Discharger treats and produces at the Michelson WRP an average monthly flow rate of 20 MGD of tertiary-2.2 quality recycled water that is delivered to use sites and/or stored and during the rainy season, when demand

is low, the recycled water can also be supplied to the Orange County Water District's (OCWD's) Green Acres Project (GAP). Under very low demand circumstances and when storage reservoirs are at almost maximum operating capacity, the Discharger currently has the ability to re-route most of the raw sewage flow that is treated at the Michelson WRP to OCSD's wastewater treatment plants (WWTPs) for treatment/disposal and/or sends recycled water to OCSD Plant No. 2 for disposal through their Ocean outfall.

10. However, during a large rain storm event that occurred during February 2017, OCSD's WWTPs experienced very high influent peak flows that significantly stressed the hydraulic capacity of their headworks; which could have been alleviated by reducing the raw sewage influent flow received from controllable sources such as IRWD's sanitary sewer system. As a result, as another large rain storm event had been forecasted for the Orange County area during February 2017 and to avoid a similar headworks' hydraulic overload episode OCSD staff requested that IRWD temporarily cease sending raw sewage flows, normally treated at the Michelson WRP, to OCSD's WWTPs. At the time, IRWD was running short of recycled water storage capacity, recycled water demand was very low, and could only discharge less than half of their recycled water flow through OCSD's Ocean outfall. Although the magnitude of the forecasted rainstorm event did not materialize and IRWD resumed their discharge of sewage to OCSD's WWTPs after the rain storm event ceased, the possibility exists that during an extended and intense rainy season this type of episodes might reoccur. Additionally, when the Discharger's solids handling and biosolids facility, currently under construction, comes online, IRWD's ability to divert sewage away from Michelson WRP to OCSD WWTPs will be limited. Therefore, it is warranted to allow IRWD to discharge, during brief periods and under the emergency conditions explained above, to Reach 1 of San Diego Creek to avoid an unauthorized release of raw sewage from the IRWD's and/or OCSD's sanitary sewer system and/or uncontrolled discharge of recycled water from IRWD's recycled water system (amongst other potential consequences). IRWD is pursuing the expansion of its recycled water storage capacity at Syphon Reservoir to almost 10 times the current reservoir capacity. IRWD is working towards developing an emergency discharge point to Reach 1 of the Santa Ana River for tertiary treated effluent. Before IRWD discharges to Reach 1 of San Diego Creek, the Discharger must meet several conditions and notify this Regional Water Board in advance regarding their intent to discharge.
11. It is appropriate to amend Order No. R8-2015-0024 to modify the reclamation specifications that apply to Discharge Point 008 (DP 008) and incorporate discharge specifications that are included in San Diego Regional Water Board's Order 97-52 for discharges of recycled water into use areas located within the jurisdictional boundaries of the San Diego Regional Water Board and regulate them under Order No. R8-2015-0024. This amendment adds Attachments "P" and "Q" to Order No. R8-2015-0024. In addition, it is appropriate to amend Order No. R8-2015-0024 to modify and/or add pertinent requirements and

provisions to regulate the discharge of tertiary treated effluent through Discharge Point Storm-007 to Reach 1 of San Diego Creek under extreme emergency conditions.

12. In accordance with Water Code Section 13389, amending the waste discharge requirements for this discharge is exempt from those provisions of the California Environmental Quality Act (CEQA) contained in Chapter 3 (commencing with Section 21100), Division 13 of the Public Resources Code.
13. The Board has notified the discharger and other interested agencies and persons of its intent to amend waste discharge requirements for the discharge and has provided them with an opportunity to submit their written views and recommendations.
14. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. R8-2015-0024 be amended as follows:

1. Order No. R8-2015-0024, page 1, replace Table 2., as follows:

Table 2. Discharge Locations

Discharge Point	Effluent Description	Discharge Point Latitude	Discharge Point Longitude	Receiving Water
001	Recycled water from MWRP to distribution system	33°39'50"N	117°50'17"W	Irvine GMZ
002	Recycled water from MWRP to San Joaquin Reservoir	33°37'12"N	117°50'39"W	Irvine GMZ
003	Tertiary treated effluent from MWRP	33°43'37"N	117°44'27"W	Rattlesnake Reservoir & Irvine GMZ
004	Tertiary treated effluent from MWRP	33°38'55"N	117°47'47"W	Sand Canyon Reservoir & Irvine GMZ
007	Tertiary treated effluent from MWRP	33°42'36"N	117°43'56"W	Syphon Reservoir & Irvine GMZ
008	Recycled water from LAWRP and/or MWRP to distribution system	33°38'16"N	117°42'50"W	Irvine GMZ & Aliso Creek Hydrologic Sub-area (Per San Diego Regional Board's Basin Plan)
Storm-007	Pumped stormwater or tertiary treated effluent (emergency) from MWRP	33°39'47"N	117°50'16"W	San Diego Creek Reach 1 tributary to Upper Newport Bay
Storm-008	Pumped stormwater (emergency) from MWRP	33°39'51"N	117°50'27"W	San Joaquin Freshwater Marsh tributary to Upper Newport Bay

2. Order No. R8-2015-0024, List of Attachments, page 4, replace the List of Attachments, as follows:

Attachment A – Definitions	A-1
Attachment B – Map.....	B-1
Attachment C – Flow Schematic	C-1
Attachment D – Standard Provisions.....	D-1
Attachment E – Monitoring and Reporting Program	E-1
Attachment F – Fact Sheet.....	F-1
Attachment G – EPA PPL Table	G-1
Attachment H – ML Table.....	H-1
Attachment I – Triggers for Monitoring Priority Pollutants	I-1
Attachment J – Stormwater Pollution Prevention Plan	J-1
Attachment K – Stormwater Monitoring and Reporting Requirements	K-1
Attachment L – Acceptance Conditions for UV Disinfection System	L-1
Attachment M – Reasonable Potential Analysis	M-1
Attachment N – Water Quality Based Effluent Limits Calculation.....	N-1
Attachment O – Total Ammonia-Nitrogen Limit Calculation	O-1
Attachment P – San Diego Regional Board’s Designation Agreement Letter	P-1
Attachment Q – Santa Ana Regional Board’s Designation Agreement Letter	Q-1

3. Order No. R8-2015-0024, Findings, page 6, add paragraph 3. and 4. to Section II.A., as follows:

3. The San Diego Regional Water Quality Control Board (San Diego Regional Water Board) regulates the Discharger’s production and distribution of recycled water and use areas, within its region’s Aliso Creek Hydrologic Sub-area, under Order 97-52, “*Waste Discharge and Water Recycling Requirements for the Production and Purveyance of Recycled Water by Member Agencies of the South Orange County Reclamation Authority*,” which is a regional water recycling permit. In a letter dated March 30, 2016, the Discharger requested to this Regional Water Board and the San Diego Regional Water Board the designation of this Regional Water Board as the sole Regional Water Board for the regulation of the treatment, distribution and use areas of recycled water produced at the Discharger’s water recycling plants. Including the distribution of recycled water to use areas located within the boundaries of the San Diego Regional Water Board in the City of Lake Forest. The Discharger made the regional water board designation request pursuant to Section 13228(a) of the California Water Code. The San Diego and this Regional Water Boards agreed with IRWD’s request to designate this Regional Water Board as the sole Regional Water Board for regulating the IRWD’s Water Recycling Plants. The San Diego and this Regional Water Boards memorialized the designation agreement in separate letters addressed to the Discharger, which were dated July 22, 2016 and September 19, 2016 respectively. Copies of these designation agreement letters are included in this Order as Attachment P and Q.

4. On July 18, 2017 the Discharger submitted a Report of Waste Discharge requesting authorization to temporarily discharge tertiary treated and disinfected effluent, under emergency conditions and as last resort for disposal, to Reach 1 of San Diego Creek through Discharge Point Storm-007. The Discharger submitted several documents in support of the propose discharge and in response to information request from Regional Water Board staff to demonstrate that the proposed discharge complies with The Water Quality Control Policy for the Enclosed Bays and Estuaries of California, as amended by SWRCB's Resolution No. 95-84 on November 16, 1995, does not impact Reach 1 of San Diego Creek and Newport Bay and complies with the conditions of the nutrient Total Maximum Daily Load (TMDL) and other TMDLs (fecal coliform, sediment and toxic pollutants) established for the San Diego Creek/Newport Bay watershed. Regional Water Board staff requested and received additional information from the Discharger until June 6, 2018.
4. Order No. R8-2015-0024, Findings, page 6, replace Section II.B., as follows:
 - B. Facility Description.** The Discharger owns and operates the MWRP and its wastewater treatment systems consist of preliminary, primary, secondary, and tertiary treatment. Recycled water is directly delivered from Discharge Point 001 (DP-001) to customers for irrigation and other recycled water reuse, except for indirect or direct potable water reuse. Recycled water from MWRP is discharged through DP-003, DP-004 and DP-007 to Rattlesnake Reservoir, Sand Canyon Reservoir, and Syphon Reservoir respectively, for storage and subsequent distribution. These three reservoirs are considered waters of the United States. Under emergency conditions, the Discharger may discharge stormwater or tertiary treated effluent through Discharge Point Storm-007 to Reach 1 of San Diego Creek and stormwater, through Discharge Point Storm-008, to San Joaquin Freshwater Marsh; which are tributary to Newport Bay. The recycled water use areas overlie the Irvine Groundwater Management Zone and the San Diego Regional Water Board's Aliso Creek Hydrologic Sub-area. Also, recycled water can be routed to the Orange County Sanitation District (OCSD) for disposal through their ocean outfall and/or Orange County Water District's (OCWD) Green Acres Project's (GAP's) recycled water distribution system. Both OCSD's ocean outfall discharge and OCWD's GAP are regulated by this Regional Water Board under separate permits.

Because of high groundwater elevation at the MWRP site, dewatering of the shallow groundwater zone is necessary to protect in-ground facilities. The area is dewatered through a network of shallow zone wells with two separate discharge points. These discharge points discharge either into Reach 1 of San Diego Creek or into the San Joaquin Freshwater Marsh. Reach 1 of San Diego Creek and San Joaquin Freshwater Marsh are tributaries to San Diego Creek and waters of the U.S. Order No. R8-2007-0003 also regulated the discharges from these dewatering wells; however, this Order does not regulate

the groundwater dewatering discharges from the MWRP (see Attachment F of this Order for details).

Also, the Discharger owns and operates the LAWRP and its wastewater treatment processes include preliminary, secondary, and tertiary treatment. Furthermore, the Discharger distributes from LAWRP Tertiary-2.2 recycled water, which is produced at the Facilities, through DP-008 to use areas located within the boundaries of this Regional Water Board and the San Diego Regional Water Board, in the City of Lake Forest, for irrigation and other recycled water reuse, except groundwater recharge. Alternatively, the LAWRP may discharge undisinfected secondary treated effluent to the Pacific Ocean through the Aliso Creek ocean outfall operated by SOCWA. The discharge to the Pacific Ocean is regulated by the San Diego Regional Water Board through Order No. R9-2012-0013 issued to SOCWA.

5. Order No. R8-2015-0024, Findings, page 7, replace Section II.C., as follows:

C. Legal Authorities. This Order is issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. Environmental Protection Agency (USEPA) and Chapter 5.5, Division 7 of the California Water Code (commencing with Section 13370). It shall serve as a NPDES permit for point source discharges from the MWRP to surface waters. This Order also serves as Waste Discharge Requirements (WDRs) pursuant to Article 4, Chapter 4, Division 7 of the California Water Code (commencing with section 13260) and as a master reclamation permit pursuant to Section 13523.1 of Article 4, Chapter 7, Division 7 of the California Water Code. In addition, this Order includes reclamation specification for the use of recycled water produced at IRWD's Facilities at use areas located within the jurisdictional boundaries of the San Diego Regional Water Board, in the City of Lake Forest, in accordance with a Regional Water Board designation request made by IRWD pursuant to Section 13228(a) of the California Water Code. The San Diego and this Regional Water Boards agreed with IRWD's request to designate this Regional Water Board as the sole Regional Water Board for regulating the IRWD's Water Recycling Plants and this designation will become effective once the San Diego Regional Water Board amends Order 97-52 or adopts a new master reclamation permit for the South Orange County Wastewater Authority that excludes IRWD.

6. Order No. R8-2015-0024, Findings, page 7, replace Section II.E., as follows:

E. California Environmental Quality Act (CEQA). Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of CEQA, Public Resources Code section 21000 et seq. (County of Los Angeles v. California State Water Resources Control Board (2006) 143 Cal.App.4th 985, mod. (Nov. 6, 2006, B184034) 50 Cal.Rptr.3d 619, 632-636). This action also involves the re-issuance of waste discharge requirements for existing

Facilities that discharge treated wastewater to land and as such, is exempt from the provisions of the California Environmental Quality Act (commencing with Section 21100) in that the activity is exempt pursuant to Title 14 of the California Code of Regulations Section 15301. The Discharger completed the MWRP capacity expansion project Environmental Impact report, SCH 2005051174 on December 28, 2005. The EIR was certified on February 27, 2006. The CEQA documents for discharge location DP-007, Syphon Reservoir, were completed on December 10, 2009 and a negative declaration was certified on December 14, 2009. The Final Supplemental Environmental Impact Report (SEIR) No. 1 for the MWRP Phase 2 and 3 Capacity Expansion Project, Biosolids Handling Component, SCH 2011031091, was certified on October 23, 2012. The SEIR No. 1 covered the CEQA process for the Biosolids Handling and Energy Recovery Facility and includes the mitigation of potential odor sources with an adequate odor control system, monitoring for presence of endangered species and/or installation of noise control measures during nesting season, and considers measures to protect human health and safety during construction. The CEQA document, Addendum No. 4 to the Michelson Water Reclamation Plant Phase 2 and 3 Capacity Expansion Project (Final EIR), addresses the temporary emergency discharge of tertiary treated effluent through Discharge Point Storm-007 to Reach 1 of San Diego Creek and was completed on March 2018 and a negative declaration was certified by IRWD on March 26, 2018. Regional Board staff has reviewed the above-referenced CEQA documents and finds that if the Discharger complies with the requirements specified in this Order, the discharges regulated under this Order should not have any significant impact on the water quality of the receiving waters.

7. Order No. R8-2015-0024, Findings, page 8, 9 and 10, replace Section II.H. and Table 5., as follows:

H. Water Quality Control Plans. The Regional Water Board adopted a revised Water Quality Control Plan for the Santa Ana Region (hereinafter Basin Plan) that became effective on January 24, 1995. The Basin Plan designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters in the Santa Ana Region addressed through the plan. More recently, the Basin Plan was amended significantly to incorporate revised boundaries for groundwater subbasins, now termed “management zones”, new nitrate-nitrogen and TDS objectives for the new management zones, and new nitrogen and TDS management strategies applicable to both surface and ground waters.

This Basin Plan Amendment was adopted by the Regional Water Board on January 22, 2004. The State Water Resources Control Board (State Water Board) and Office of Administrative Law (OAL) approved the Amendment on September 30, 2004 and December 23, 2004, respectively. EPA approved the

surface water standards components of the N/TDS Amendment on June 20, 2007. The San Diego Regional Water Board adopted the Water Quality Control Plan for their region on September 8, 1994 and subsequently approved by the State Water Resources Control Board (SWRCB) on December 13, 1994. Subsequent revisions to the San Diego Regional Water Board's Basin Plan have also been adopted by their board and approved by the SWRCB. The San Diego Regional Water Board's Basin Plan contains beneficial uses and water quality objectives. The reclamation specifications for recycled water use, within the Aliso Creek Hydrologic Sub-area, of this Order are consistent with the San Diego Regional Water Board's Basin Plan.

In addition, the Basin Plan implements State Water Resources Control Board (State Water Board) Resolution No. 88-63, which established State policy that all waters should be considered suitable or potentially suitable for municipal or domestic supply (i.e., designated MUN), unless certain exception criteria apply. Pursuant to these criteria, the Regional Board exempted certain waters, including Rattlesnake, Syphon, and Sand Canyon Reservoirs, San Diego Creek, and Newport Bay, from the MUN designation. The Aliso Creek Hydrologic Sub-area is not exempted from the MUN designation under the San Diego Regional Water Board's Basin Plan.

Table 5 Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
001-008, Storm-007 & 008	Irvine Groundwater Management Zone	<u>Present or Potential:</u> Municipal and domestic supply, agricultural supply, industrial service supply, and industrial process supply.
003	Rattlesnake Reservoir	<u>Present or Potential:</u> Agricultural supply, water contact recreation, non-contact water recreation, warm freshwater habitat, wildlife habitat Excepted from Municipal and Domestic Supply
004	Sand Canyon Reservoir	<u>Present or Potential:</u> Agricultural supply, water contact recreation, non-contact water recreation, warm freshwater habitat, wildlife habitat Excepted from Municipal and Domestic Supply
007	Syphon Reservoir	<u>Present or Potential:</u> Agricultural supply, water contact recreation, non-contact water recreation, warm freshwater habitat, wildlife habitat Excepted from Municipal and Domestic Supply
001 & 008	Aliso Creek Hydrologic Sub-area (Per San Diego Regional Board's Basin Plan)	<u>Present or Potential:</u> Municipal and domestic supply and agricultural supply.

Table 5 Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
Storm-007	San Diego Creek, Reach 1	<u>Present or Potential:</u> Water contact recreation, non-contact water recreation, warm freshwater habitat, and wildlife habitat Excepted from Municipal and Domestic Supply
Storm-008	San Joaquin Freshwater Marsh	<u>Present or Potential:</u> Water contact recreation; non-contact water recreation; warm freshwater habitat; preservation of biological habitats of special significance; rare, threatened or endangered species; and wildlife habitat Excepted from Municipal and Domestic Supply
Storm-007 & Storm-008	Newport Bay, Upper	<u>Present or Potential:</u> Navigation, Water contact recreation, Non-contact water recreation, Commercial and sport fishing, Preservation of biological habitats of special significance, Wildlife habitat, Rare, threatened or endangered species, Spawning, reproduction, and development, Marine habitat, Shellfish harvesting, and Estuarine habitat. Excepted from Municipal and Domestic Supply

Requirements of this Order implement the Basin Plans of this Regional Water Board and the San Diego Regional Water Board.

8. Order No. R8-2015-0024, Findings, page 13, replace Section II.V., as follows:

V. TMDLs. On October 11, 2011, the U. S. EPA issued its final decision regarding the State's 2010 Integrated Clean Water Act Sections 303(d) and 305(b) list of impaired water bodies. This list includes Reaches 1 and 2 of San Diego Creek and Upper and Lower Newport Bay. One or more of these water bodies were determined to be impaired by one or more pollutants, including: fecal coliform, nutrients, metals and pesticides. Nutrient, fecal coliform, sediment, and toxic pollutants TMDLs have been established for the San Diego Creek/Newport Bay watershed. There is no discharge of treated wastewater or storm water to these impaired waterbodies under normal conditions (overflow from Sand Canyon is allowed only when the flow in San Diego Creek at Campus Drive is above 50 cubic feet per second that is under extreme storm conditions). In addition, under emergency conditions and on a temporary basis the Discharger can discharge tertiary treated effluent to Reach 1 of San Diego Creek, which is tributary to Newport Bay, provided that several conditions are met and the total nitrogen and total phosphorous loads that are discharged and not excluded by the nutrient TMDL are offset at a ratio of 1.5:1. This Order does implement relevant TMDL requirements (see Attachment F).

9. Order No. R8-2015-0024, Discharge Prohibitions, page 14, replace Section III.F.4., as follows:

4. For discharges of recycled water from Sand Canyon Reservoir to San Diego Creek or its tributaries when the flow in San Diego Creek at Campus Drive is less than 50 cubic feet per second, the Discharger shall have an approved offset program to offset total nitrogen loads, discharged in excess of 1 mg/L total nitrogen concentration, and any total phosphorous loads discharged.
10. Order No. R8-2015-0024, Effluent Limitations and Discharge Specifications, page 14 and 15, replace title and paragraph of Section IV.A., title of Section IV.A.1. and title of Table 6., as follows:

A. Tertiary Treated Effluent Limitations – Discharge Points 003, 004, 007 & Storm-007

Unless otherwise specifically specified hereinafter, compliance with the following effluent limitations is measured at monitoring locations M-003A, M-004A, M-007, and M-008 as described in the attached MRP (Attachment E).

1. Final Effluent Limitations – Discharge Point 003, 004, 007 & Storm-007

- a. The Discharge shall maintain compliance with the following effluent limitations:

Table 6. Effluent Limitations at DPs 003, 004, 007 & Storm-007

11. Order No. R8-2015-0024, Effluent Limitations and Discharge Specifications, page 14, replace footnote No. 4, as follows:

Compliance with this limitation is to be achieved at M-003B, M-004B, M-007 and M-008 as described in the attached MRP (Attachment E.). Also, for M-008 see Section VII.O – Compliance Determination.

12. Order No. R8-2015-0024, Effluent Limitations and Discharge Specifications, page 15, replace Section IV.A.1.c., as follows:

- c. **TDS Limitations** – For DPs 003, 004 and 007 the 12-month flow weighted running average TDS constituent concentrations shall not exceed 720 mg/L and for Storm-007 the 12-month flow weighted running average TDS constituent concentrations shall not exceed 1,500 mg/L.

13. Order No. R8-2015-0024, Effluent Limitations and Discharge Specifications, page 16, add Section IV.A.1.f. and g., as follows:

- f. The Discharger shall offset the mass loads (in lbs/rainfall year) of Total Nitrogen (TN) and Total Phosphorous (TP) discharged, through Storm-007, to Reach 1 of San Diego Creek at a ratio 1.5:1 in accordance with the following conditions:
 - (1) When San Diego Creek flow is less than 50 cfs at Campus Drive TN mass loads discharged must be offset.
 - (2) TP mass loads discharged must be offset under any flow condition in San Diego Creek at Campus Drive.
- g. The Discharger shall quantify and report the TN and TP offset credits, in lbs/year, for every rainfall year (October-September) starting with the rainfall year of October 2017 through September 2018. The Discharger shall report the TN and TP removal credits available for offset in the annual monitoring report. The Discharger shall include in the offset report details regarding the location of TN and TP removal sites, inlet/outlet concentrations, pertinent dates, analytical methods used, equations used, map depicting the TN and TP removal locations, and load removal calculations in a tabulated format. TN and TP mass loads discharged to San Diego Creek must be offset with TN and TP mass removal credits accumulated within a three-rainfall year period that includes the rainfall year in which the discharge occurred.

14. Order No. R8-2015-0024, Effluent Limitations and Discharge Specifications, Reclamation Specifications, page 17, replace Table 7 and Section IV.C.1.b., as follows:

Table 7 Recycled Water Effluent Limitations at DPs 001, 002, & 008

Parameter	Units	Effluent Limitations		
		Average Monthly	Average Weekly	12-month running average
Biochemical Oxygen Demand 5-day @ 20°C	mg/L	20	30	- -
Total Suspended Solids	mg/L	20	30	- -
pH	standard units	6 (instantaneous minimum)	9 (instantaneous maximum)	- -
Chloride	mg/L			375
Sulfate	mg/L			375
% Sodium	%			60
Nitrate	mg/L			45
Iron	mg/L			0.3
Manganese	mg/L			0.05
MBAS	mg/L			0.5
Boron	mg/L			0.75

Table 7 Recycled Water Effluent Limitations at DPs 001, 002, & 008

Parameter	Units	Effluent Limitations		
		Average Monthly	Average Weekly	12-month running average
Aluminum	mg/L			1.0
Arsenic	mg/L			0.05
Antimony	mg/L			0.006
Barium	mg/L			1.0
Beryllium	mg/L			0.004
Cadmium	mg/L			0.005
Cyanide	mg/L			0.2
Mercury	mg/L			0.002
Nickel	mg/L			0.1
Selenium	mg/L			0.05
Thallium	mg/L			0.002

Note: The 12-month running average effluent limitations on this table apply only to recycled water use that overlies the Aliso Creek Hydrologic Sub-area within the San Diego Regional Water Board jurisdictional boundary.

- b. TDS Limitations: For recycled water use on sites overlying the Irvine Groundwater Management Zone, the 12-month flow weighted running average TDS concentration shall not exceed 910 mg/L. For recycled water use on sites overlying the Aliso Creek Hydrologic Sub-area, the 12-month running average TDS concentrations shall not exceed 1,000 mg/L.
15. Order No. R8-2015-0024, Effluent Limitations and Discharge Specifications, Reclamation Specifications, page 21, add Section IV.C.10. and Section IV.C.11., as follows:
 10. Uses of recycled water with frequent or routine application (for example: agricultural or landscape irrigation uses) shall be at agronomic rates and shall consider soil, climate, and plant demand. In addition, application of recycled water and use of fertilizers shall be at a rate that takes into consideration nutrient levels in recycled water and nutrient demand by plants. This Regional Water Board may require the Discharger to submit an Implementation or Operations and Management Plan specifying agronomic rates and nutrient application for the use area(s) and a set of reasonably practicable measures to ensure compliance with this Order. The Discharger must implement the following to ensure that recycled water and fertilizer are applied in use sites at agronomic rates:
 - a. Monitor nutrient concentrations in recycled water supplies and notify recycled water site supervisors of the nutrient concentrations of recycled water.

- b. Conduct periodic inspections of end use sites to verify the agronomic application rates of recycled water considering fertilizer application and the nutrient level of the recycled water applied to the landscape or provide monthly water budgets to each use site based on agronomic rates, provide monthly reporting to each use site of actual use compared with the agronomic water budget for each use site, and target sites with recycled water consistently above the agronomic rate for follow-up.
- 11. In lieu of meeting the recycled water effluent limit for total nitrogen of 10 mg/L (Section IV.C.1.c., Table 7), the Discharger may elect to conduct a nitrogen study to verify that the use of recycled water from IRWD's Facilities for landscape or agricultural irrigation does not cause the water quality of the underlying groundwater to exceed the groundwater quality objective for nitrate of 45 mg/L (as nitrate) nor exacerbate an existing condition that exceeds the nitrate groundwater quality objective. The nitrogen study must evaluate factors such as existing nitrogen removal achieved at the plant, need for additional treatment processes to remove nitrogen, fate and transport of nitrogen in the groundwater, groundwater monitoring, application of recycled water and fertilizer at agronomic rates, nitrogen uptake by turf grass or other vegetation, and other best management practices. A workplan for the nitrogen study, if needed, must be received at our Santa Ana Regional Water Board's office by June 15, 2019, and must identify proposed tasks and milestones for completing the nitrogen study, and a schedule for completing the study activities. A nitrogen study report documenting the results of the study must be received at this Regional Water Board by March 15, 2020. For a valid reason, the Executive Officer may grant additional time to submit the workplan and/or report for the nitrogen study.
- 16. Order No. R8-2015-0024, Provisions, page 28, add Section VI.C.2.e., as follows:
 - e. By December 31, 2018, the Discharger shall submit for approval by the Executive Officer, a TN and TP Offset plan that details the manner in which the Discharger will gather all pertinent information to document the removal credits of TN and TP loads in the NTS and San Joaquin Freshwater Marsh managed by IRWD and the reporting of the data as required in this Order. For a valid reason, the Executive Officer may grant additional time to submit the TN and TP Offset plan.
- 17. Order No. R8-2015-0024, Provisions, Other Special Provisions, page 33, add Section VI.C.6.b through d., as follows:
 - b. Before tertiary treated effluent is discharged to Reach 1 of San Diego Creek, through Discharge Point (DP) Storm-007, the Discharger shall give this Regional Water Board and the California Department of Fish and

Wildlife (Carlsbad Office) at least 48-hour advance notice regarding their intent to discharge and fulfill the following conditions:

- (1) Exhaust its recycled water storage capacity;
 - (2) Stop importing water purchased from the Metropolitan Water District to augment its recycled water system;
 - (3) Shutoff all irrigation water wells providing water to its recycled water system;
 - (4) Stop producing recycled water at Los Alisos Water Recycling Plant;
 - (5) Send recycled water to the Orange County Water District (OCWD) through the Green Acres Project;
 - (6) Dispose of recycled water, to the maximum extent possible, through the Orange County Sanitation District's (OCSD's) ocean outfall;
 - (7) If possible, shutoff all Department of Justice wells that deliver water into the recycled water system, while still complying with the Record of Decision with the Department of the Navy pertaining to the operation of these wells;
 - (8) Send all allowable recycled water (subject to physical conveyance capacities) to South Orange County Wastewater Authority's ocean outfall;
 - (9) Confirm and document that OCWD and OCSD cannot take additional recycled water flow; and
 - (10) If a diversion of sewage is feasible from Michelson WRP to OCSD, confirm that OCSD cannot receive raw sewage flow from IRWD MWRP's service area.
- c. If a diversion of sewage is feasible from Michelson WRP to OCSD, IRWD shall maintain real-time communication and coordination with OCSD staff so that IRWD can resume, as-soon-as-possible, the disposal of raw sewage from IRWD MWRP's service area through OCSD treatment plants and discontinue the discharge of tertiary treated effluent to Reach 1 of San Diego Creek through DP Storm-007.
- d. The Discharger must develop and implement a rainy season discharge contingency plan that includes details describing the measures that IRWD will take to operate the Michelson WRP in an enhanced nutrient removal mode to produce recycled water with an average total nitrogen and total phosphorous concentrations of 10 mg/L and 0.3 mg/L respectively, as operational goals, and comply with the total chlorine residual effluent limits (amongst other parameters) when discharging tertiary treated effluent through DP Storm-007. This plan must include pertinent details regarding the operational strategies, additional equipment and resources, chemicals, timing of implementation, testing, power loss backup equipment, planning and preparation, and OCSD/OCWD/IRWD staff coordination and contact list (emails, phone numbers, and functions). The rainy season discharge

contingency plan must be made available to Regional Water Board staff upon request.

18. Order No. R8-2015-0024, Compliance Determination, page 35, replace Section VII.I., as follows:

I. Total Chlorine Residual Limitation (TCR)

For Rattlesnake and Sand Canyon Reservoirs the compliance determination for total chlorine residual shall be determined at the boundary of the zone of initial dilution approved by the Regional Board Executive Officer and for Syphon Reservoir is determined at the discharge point. For tertiary treated effluent discharged through Discharge Point Storm-007 the compliance determinations for total chlorine residual shall be based on 99% compliance. To determine 99% compliance with the effluent limitation for total chlorine residual, the following conditions shall be satisfied:

1. For TCR Limit specified in Section IV.A.1.:
 - a. The total time during which the total chlorine residual values are above 0.1 mg/L (instantaneous maximum value) shall not exceed 7 hours and 26 minutes in any calendar month;
 - b. No individual excursion from 0.1 mg/L value shall exceed 5 minutes; and
 - c. No individual excursion shall exceed 5.0 mg/L.

19. Attachment C, Flow Schematic, page C-1, replace page C-1 of Attachment C with page C-1 that is attached to this Order.

20. Attachment E, Monitoring Locations, page E-7, replace Table 2, as follows:

Table 2 Monitoring Station Locations

Discharge Point Name	Monitoring Location Name	Monitoring Location Description	Latitude and Longitude
--	M-INF 1	Influent distribution channel at MWRP	33°39'48" & 117°50'19"
	M-INF 2	Influent channel at LAWRP	33°38'16" & 117°42'54"
001	REC-001	Effluent from MWRP to recycling system	33°39'50" & 117°50'17"
002	REC-002	Effluent to San Joaquin Reservoir	33°37'12" & 117°50'39"
003	M-003A	Effluent to Rattlesnake Reservoir	33°43'37" & 117°44'43"
003	M-003B	Rattlesnake Reservoir ZID boundary	33°43'37" & 117°44'27"
004	M-004A	Effluent to Sand Canyon Reservoir	33°38'55" & 117°47'47"
004	M-004B	Sand Canyon Reservoir ZID boundary	33°38'50" & 117°47'50"
007	M-007	Effluent to Syphon Reservoir	33°42'36" & 117°43'56"
008	REC-003	Effluent from LAWRP to recycling system	33°38'16" & 117°42'50"
Storm-007	M-008	Emergency stormwater overflow or tertiary treated effluent discharge to	33°39'47" & 117°50'16"

Table 2 Monitoring Station Locations

Discharge Point Name	Monitoring Location Name	Monitoring Location Description	Latitude and Longitude
		Reach 1 of San Diego Creek	
Storm-008	M-009	Emergency stormwater overflow to San Joaquin Marsh	33°39'51" & 117°50'27"
	R-001	Receiving surface water—Rattlesnake Reservoir 3 location spatial composite	33°43'46" & 117°44'12" 33°43'42" & 117°44'23" 33°43'41" & 117°44'29"
	R-002	Receiving surface water—Sand Canyon Reservoir location 3 spatial composite	33°38'38" & 117°47'46" 33°38'48" & 117°47'48" 33°38'51" & 117°47'47"
	R-003	Receiving surface water—Syphon Reservoir location 3 spatial composite	33°43'39" & 117°44'28" 33°43'43" & 117°44'20" 33°43'46" & 117°44'13"
	R-004U	Upstream monitoring location in San Diego Creek, 100 meters upstream from Discharge Point Storm-007	33°39'48" & 117°50'15"
	R-004D	Downstream monitoring location in San Diego Creek, within 500 meters downstream from Discharge Point Storm-007	33°39'43" & 117°50'20"

21. Attachment E, Effluent Monitoring Requirements to Surface Waters, page E-9 and E-10, replace the title of Section IV.A., paragraph IV.A.1., and title of Table 4, as follows:

A. Effluent Monitoring Locations M-003, M-004, M-007 & M-008

- The Discharger shall monitor tertiary treated effluent for DP-003, DP-004, DP-007 and Storm-007 at Monitoring Locations M-003A, M-004A, M-007 and M-008, respectively, as follows.

Table 4 Tertiary Effluent Monitoring M-003A, M-004A, M-007 & M-008

22. Attachment E, Effluent Monitoring Requirements to Surface Waters, page E-12, replace Section IV.A.2. and Table 5, as follows:

- The Discharger shall monitor tertiary treated effluent for DP 003, DP 004, DP 007 and Storm-007 at Monitoring Locations M-003B, M-004B, M-007 and M-008, respectively, for Total Residual Chlorine and Total Dissolved Solids as follows.

Table 5 Tertiary Effluent Monitoring M-003B, M-004B, M-007 & M-008*

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method and Minimum Level, units, respectively
Total Residual Chlorine	mg/L	Grab	Weekly	See Section I.A.3. above of this MRP
Total Residual Chlorine*	mg/L	Recorder	Continuous	--
Total Dissolved Solids*	mg/L	Composite	Monthly	See Section I.A.3. above of this MRP

Note: * = Apply only at M-008 when discharging tertiary treated effluent.

23. Attachment E, Effluent Monitoring Requirements to Surface Waters, page E-13, replace Sections IV.A.5. and IV.A.7. as follows:

5. If there are no discharges of tertiary treated effluent to DP 003, DP 004, DP 007 and Storm-007, the Discharger is not required to monitor for all the constituents in Table 4 and Table 5, above. However, the Discharger shall record on a permanent log on a daily basis when there is no discharge and report this information together with the monthly report.
7. The Discharger may demonstrate compliance with the Effluent Limitations (except for Total Residual Chlorine and Total Dissolved Solids at M-008), Discharge Specifications, and Monitoring and Reporting Requirements in this Order for tertiary treated effluent discharges at Discharge Locations DP 002, DP 003, DP 004, DP 007 and Storm-007 by monitoring at Discharge Location DP 001 (monitoring location REC-001).

24. Attachment E, Whole Effluent Toxicity Testing Requirements, page E-13, replace the title of Section V.A., as follows:

A. Toxicity Monitoring Requirements at M-003A, M-004A, M-007 and M-008

25. Attachment E, Recycled Water Monitoring Requirements, page E-16, replace Table 6, as follows:

Table 6 Reclamation Monitoring at REC-001, REC-002, & REC-003

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Flow	mgd	Recorder/Totalizer	Continuous	---
pH	Standard units	Recorder/Totalizer	Continuous	--
CT ¹⁴	mg/L-min	Recorder	Continuous	See Section I.A.3., above, of this MRP
Turbidity	NTU	Recorder	Continuous	"
Coliform Organisms	MPN per 100 mL	Grab	Daily	"
BOD ₅	mg/L	Composite	Weekly	"
Total Suspended Solids	mg/L	Composite	Weekly	"

TDS	mg/L	Composite	Monthly	"
Chloride	mg/L	Composite	Quarterly	"
Percent Sodium	%	Composite	Quarterly	"
Sulfate	mg/L	Composite	Quarterly	"
Iron	mg/L	Composite	Quarterly	"
Manganese	mg/L	Composite	Quarterly	"
MBAS	mg/L	Composite	Quarterly	"
Boron	mg/L	Composite	Quarterly	"
Aluminum	mg/L	Composite	Annually	"
Barium	mg/L	Composite	Annually	"
Nitrate-Nitrogen	mg/L	Composite	Quarterly	"
Total Nitrogen	mg/L	Composite	Quarterly	"
Priority Pollutants	mg/L	Composite	Annually	"

26. Attachment E, Recycled Water Monitoring Requirements, page E-16, add Section VII.A.2., as follows:

2. The Discharger shall increase the sampling frequency from weekly to daily, from quarterly to monthly, and from annually to quarterly for any constituent that exceeds the recycled water effluent limitations specified in Table 7 of this Order. To return to the monitoring frequency specified, the Discharger shall request and receive approval from the Regional Water Board's Executive Officer or designee after maintaining compliance with the effluent limitations in question for at least three consecutive monitoring periods.

27. Attachment E, Recycled Water Monitoring Requirements, page E-16, replace Section VII.B., as follows:

B. Monitoring Users

The Discharger shall record the information described below and shall submit annually a summary report of the recorded information by groundwater management zone to the Regional Water Board.

1. Use site information summary that includes the following: a) Name and address of use site; b) Owner of the site; c) Name and phone number of the onsite supervisor; d) Groundwater management zones underlying the use sites; e) Annual volume of recycled water delivered; f) Total area (in acres) of each landscape irrigation site; g) The amount of nitrogen (in pounds per acre per year) applied in recycled water on each landscape irrigation site (required if the Discharger elects to implement a nitrogen study per Section IV.C.11. of the Order); and h) Type of use (e.g. irrigation, industrial, etc.).
2. Recycled water use site inspection summary that includes the number of inspections conducted by the Discharger and number of violations, if any.

3. Summary of compliance status and enforcement actions taken by the Discharger for recycled water users known to be in violation of its rules and regulations, which include a description of violations (cause, duration, and impact), measures taken to correct and eliminate recurrence of a violation or threatened violation, and compliance schedules issued to use sites.
28. Attachment E, Receiving Water Monitoring Requirements – Surface Water and Groundwater, page E-17, replace the paragraph of Section VIII.A.1. and title of Table 7, as follows:
1. The Discharger shall monitor receiving water at R-001, R-002, R-003, R-004U & R-004D as follows:

Table 7 Receiving Water Monitoring Requirements at R-001, R-002, R-003, R-004U & R-004D

29. Attachment F, List of Tables, page F-2 and F-3, replace the List of Tables, as follows:

List of Tables

Table 1.	Facility Information	22
Table 2.	Facilities' Treatment Processes	Error!
Bookmark not defined.		
Table 3.	Recycled Water Produced	Error!
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Table 4.	Closed Tanks	Error!
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Table 5.	Open Reservoirs	Error!
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Table 6.	Treated Water Sent to OCWD and OCSD	Error!
Bookmark not defined.		
Table 7.	Summary of Discharge Points and Receiving Waters	25
Table 8.	Historic Effluent Monitoring Data at DPs 003, 004 and 007	Error!
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Table 13.	MWRP Effluent Monitoring Data, µg/L	Error!
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Table 14.	RPA Evaluation for DPs 003, 004, & 007	Error!
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Table 16.	Summary of Water Quality-based Final Effluent Limitations for Discharge Points 003, 004, & 007	34
Table 17	Recycled Water Effluent Limitations at DPs 001, 002, & 008	F-34
Table 18.	Stormwater Discharge Limits at Storm-007 & Storm-008	Error!
	Bookmark not defined.	

30. Attachment F, Permit Information, page F-4, replace Table 1, as follows:

Table 1. Facility Information

WDID	8 302006002	
Discharger/Operator	Irvine Ranch Water District	
Names of Facilities	Michelson Water Recycling Plant	Los Alisos Water Recycling Plant
Addresses	3512 Michelson Drive	22312 Muirlands Boulevard
	Irvine, CA 92612	Lake Forest, CA 92630
	Orange County	
Facility Contact, Title and Phone	Gaspar Garza, Operations Manager, Phone: (949) 453-5826	
Authorized Person to Sign and Submit Reports	Jose Zepeda, Director of Recycling Operations, Phone: (949) 453-5572	
Mailing Address	15600 Sand Canyon Avenue, Irvine, CA 92618	
Billing Address	P.O. BOX 57000, Irvine, CA 92619-7000	
Type of Facility	POTW	
Major or Minor Facility	Major	
Threat to Water Quality	1	
Complexity	A	
Pretreatment Program	Y	
Reclamation Requirements	Producer/User	
Facility Permitted Flow	28 million gallons per day (mgd)	7.5 mgd - secondary treatment and 5.5 mgd – tertiary treatment
Type of Permitted Flows	NPDES	WDR (Recycled Water only)
Facility Design Flow	28 mgd	7.5 mgd - secondary treatment and 5.5 mgd – tertiary treatment
Watershed	San Diego Creek/Newport Bay watershed	San Diego Creek/Newport Bay watershed
Receiving Water	Irvine Groundwater Management Zone. Aliso Creek Hydrologic Sub-area (Per San Diego Regional Board's Basin Plan) Surface waters: Rattlesnake, Syphon, and Sand Canyon Reservoirs, Reach 1 of San Diego Creek, San Joaquin Freshwater Marsh and Upper Newport Bay.	
Receiving Water Type	Inland Surface Water, Estuary and Groundwater	

31. Attachment F, Permit Information, page F-5, replace Section I.B., as follows:

- B.** The MWRP discharges wastewater to waters of the United States, including: 1) tertiary treated effluent, for recycled water storage purposes, to Sand Canyon Reservoir, Syphon Reservoir, and Rattlesnake Reservoir; 2) under emergency conditions, pumped storm water or tertiary treated effluent (as a last resort disposal alternative) to Reach 1 of San Diego Creek; and 3) under emergency conditions, pumped storm water to San Joaquin Freshwater Marsh. Both Reach 1 of San Diego Creek and the San Joaquin Freshwater Marsh are tributaries of Upper Newport Bay. The discharge from MWRP is currently regulated by Order No. R8-2007-0003, which was adopted on November 30, 2007 and expired on November 1, 2012. Order No. R8-2007-0003 was amended by Order No. R8-2008-0072 on September 5, 2008. The amendment corrected the discharge prohibition period of Sand Canyon Reservoir, modified reclamation specifications, and provided flexibility to demonstrate compliance at monitoring location REC-001 for other discharge points. Also, on June 15, 2012 Order No. R8-2007-0003 was amended by Order No. R8-2012-0004, which added Syphon Reservoir as an additional discharge point for recycled water storage in the distribution system. The terms and conditions of the current Order and its amendment have been automatically continued and remain in effect until new Waste Discharge and Water Reclamation Requirements and NPDES permit are adopted pursuant to this Order.

The LAWRP discharges tertiary treated and disinfected recycled water to its recycled water distribution system within the Lake Forest service area and/or to IRWD's Zone B. Recycled water produced at the Facilities is distributed within the City of Lake Forest to use areas located within the jurisdiction of this Regional Water Board and the San Diego Regional Water Board, however, as explained in Section I.C below, this Regional Water Board has been designated to be the sole Regional Water Board that regulates the treatment and distribution of recycled water produced at the IRWD's Facilities. When discharging to IRWD's Zone B the recycled water is not discharged to IRWD's open reservoirs that are waters of the US. On the other hand, recycled water produced at MWRP may be received at LAWRP recycled water pump station's wetwell, on a seasonal basis, for distribution within the Lake Forest service area of LAWRP. The recycled water discharge from this facility is currently regulated under Order No. 94-3 issued to the South Orange County Reclamation Authority (SOCRA) by this Regional Water Board on March 11, 1994. SOCRA changed its name in 2001 to the South Orange County Wastewater Authority (SOCWA). Also, Order No. 94-3 regulates recycled water discharges from El Toro Water District's Water Recycling Plant. Both IRWD and El Toro Water District are members of SOCWA. When recycled water from MWRP is received at LAWRP or when a portion of its secondary

treated effluent is not reclaimed, LAWRP's undisinfected secondary effluent is discharged to the Pacific Ocean through the Aliso Creek ocean outfall operated by SOCWA. The discharge to the Pacific Ocean is regulated by the San Diego Regional Water Quality Control Board through Order No. R9-2012-0013 issued to SOCWA.

32. Attachment F, Permit Information, page F-6, add a fourth and fifth paragraph at the end of Section I.C., as follows:

The San Diego Regional Water Quality Control Board (San Diego Regional Water Board) regulates the use of recycled water, produced at the Discharger's Facilities, within the Aliso Creek Hydrologic Sub-area, of its region, under Order 97-52, *Waste Discharge and Water Recycling Requirements for the Production and Purveyance of Recycled Water by Member Agencies of the South Orange County Reclamation Authority*, which is a regional water recycling permit. In a letter dated March 30, 2016, the Discharger requested to this Regional Water Board and the San Diego Regional Water Board the designation of this Regional Water Board as the sole Regional Water Board for the regulation of the treatment, distribution and use of recycled water produced at the Discharger's water recycling plants. Including the distribution of recycled water to use areas located within the boundaries of the San Diego Regional Water Board in the City of Lake Forest. The Discharger made the regional water board designation request pursuant to Section 13228(a) of the California Water Code. The San Diego and this Regional Water Boards agreed with IRWD's request to designate this Regional Water Board as the sole Regional Water Board for regulating the IRWD's Facilities. The San Diego and this Regional Water Boards memorialized the designation agreement in separate letters addressed to the Discharger, which were dated July 22, 2016 and September 19, 2016, respectively, and this designation will become effective once the San Diego Regional Water Board amends Order 97-52 or adopts a new master reclamation permit for the South Orange County Wastewater Authority that excludes IRWD's.

On July 18, 2017 the Discharger submitted a Report of Waste Discharge requesting authorization to temporarily discharge tertiary treated and disinfected effluent, under emergency conditions, to Reach 1 of San Diego Creek through Discharge Point Storm-007 (Storm-007). The Discharger submitted several documents in support of the proposed discharge and in response to information request from Regional Water Board staff to demonstrate that the proposed discharge enhances and does not impact the water quality of Reach 1 of San Diego Creek and Newport Bay and complies with the Water Quality Control Policy for the Enclosed Bays and Estuaries of California, as amended by SWRCB's Resolution No. 95-84, and the conditions of the Nutrient Total Maximum Daily Load (TMDL) and other TMDLs (fecal coliform, sediment and toxic pollutants) established for the San Diego Creek/Newport Bay watershed. Regional Water Board staff requested and received additional information from the Discharger until June 6, 2018.

33. Attachment F, Facility Description, page F-12, replace Section II.B.1, as follows:

- 1. Discharge Points for Tertiary Treated Effluent and Recycled Water Reuse**
Recycled water is delivered via DP-001 and DP-008 to the distribution system serving recycled water users.

Recycled water produced at MWRP is delivered via DP-002 to San Joaquin Reservoir for storage.

For recycled water storage purposes, tertiary treated effluent from MWRP is dechlorinated and discharged to surface waters at three points located at Rattlesnake Reservoir, Sand Canyon Reservoir, and Syphon Reservoir, designated as DP-003, DP-004, and DP-007, respectively. Tertiary treated recycled water produced at LAWRP and discharged through DP-008 is not discharged to the reservoirs that are filled with recycled water produced at MWRP. Also, under emergency conditions and as last resort for disposal, tertiary treated effluent may be temporarily discharged to Reach 1 of San Diego Creek through Storm-007, which is tributary to Upper Newport Bay.

34. Attachment F, Facility Description, page F-12, replace Table 7, as follows:

Table 7. Summary of Discharge Points and Receiving Waters

Discharge Serial No.	Latitude	Longitude	Description and Receiving Waters	Flow & Frequency
001	33°39'50"N	117°50'17"W	Recycled water from MWRP to distribution system serving recycled water users, overlying Irvine GMZ	12.3 mgd Continuous
002	33°37'12"N	117°50'39"W	Recycled water from MWRP to San Joaquin Reservoir, overlying Irvine GMZ	Variable up to 12.3 mgd
003	33°43'37"N	117°44'27"W	Tertiary treated effluent from MWRP to Rattlesnake Reservoir, overlying Irvine GMZ	Variable up to 12.3 mgd
004	33°38'55"N	117°47'47"W	Tertiary treated effluent from MWRP to Sand Canyon Reservoir, overlying Irvine GMZ	Variable up to 12.3 mgd
007	33°42'36"N	117°43'56"W	Tertiary treated effluent from MWRP to Syphon Reservoir, overlying Irvine GMZ	Variable up to 12.3 mgd
008	33°38'16"N	117°42'50"W	Recycled water from LAWRP and/or MWRP to distribution system serving recycled water users, overlying Irvine GMZ & Aliso Creek Hydrologic Sub-area (Per San Diego Regional Board's Basin Plan)	Variable up to 5.5 mgd
Storm-007	33°39'47"N	117°42'50"W	Pumped stormwater or tertiary treated effluent (emergency) from MWRP to San Diego Creek Reach 1 tributary to Upper Newport Bay	Metered

Table 7. Summary of Discharge Points and Receiving Waters

Discharge Serial No.	Latitude	Longitude	Description and Receiving Waters	Flow & Frequency
Storm-008	33°39'51"N	117°50'27"W	Pumped stormwater (emergency) from MWRP to San Joaquin Freshwater Marsh tributary to Upper Newport Bay	Unmetered

35. Attachment F, Summary of Existing Requirements and Self-Monitoring Report (SMR) Data, page F-14, replace the last paragraph of Section II.C.1., as follows:

The previous Order prohibited the direct discharge of tertiary treated recycled water to surface water, except into the three reservoirs, Rattlesnake Reservoir, Syphon Reservoir, and Sand Canyon Reservoir. However, during wet weather conditions, storm water runoff from the reservoir watershed occasionally causes overflows¹ from Sand Canyon Reservoir into San Diego Creek. The previous Order R8-2007-0003 allowed emergency discharge of water from the reservoir during heavy rainfall events. The Discharger has implemented best management practices to minimize overflows from Sand Canyon Reservoir by limiting recycled water storage during the rainy season. During the term of Order No. R8-2007-0003, Sand Canyon Reservoir has overflowed twice in 2010 and 2011 during the rainy season. This Order permits overflows from Sand Canyon Reservoir under certain storm conditions when the flow in San Diego Creek at Campus Drive is expected to be 50 cubic feet per second or higher. Under such flow conditions, the discharge should have no impact on nutrient loading to San Diego Creek or Newport Bay. Also, under emergency conditions and as last resort for disposal, tertiary treated effluent may be temporarily discharged to Reach 1 of San Diego Creek through Storm-007, which is tributary to Upper Newport Bay. The Discharger must implement a total nitrogen (TN) offset program if the discharge of recycled water, stormwater or tertiary treated effluent occurs when the flow in San Diego Creek at Campus Drive is less than 50 cubic feet per second and the TN concentration in the effluent exceeds 1 mg/L. Total phosphorous (TP) loads discharged must be offset regardless of the flow regime in San Diego Creek. IRWD operates a network of natural treatment wetlands in the San Diego Creek Watershed that remove nitrogen and phosphorus from San Diego Creek and its tributaries.

36. Attachment F, Planned Changes, page F-16, replace the last paragraph of Section II.E., as follows:

IRWD is constructing a Biosolids Handling and Energy Recovery Facility at the MWRP site, which would treat wastes from both Facilities, and includes the construction and/or installation of sludge thickening equipment, anaerobic digesters, centrifuges for sludge dewatering, sludge drying equipment, anaerobically digestible material receiving station, and microturbines for generation of electricity and heat. The microturbines would operate with biogas produced in the anaerobic digesters. IRWD anticipates completion of the Biosolids

Handling and Energy Recovery Facility by mid-2020. In addition, IRWD is pursuing the expansion of its recycled water storage capacity at Syphon Reservoir to almost 10 times the current reservoir capacity. IRWD is also working towards developing an emergency discharge point to Reach 1 of the Santa Ana River for tertiary treated effluent.

37. Attachment F, Applicable Plans, Policies, and Regulations, page F-16, replace Section III.A., as follows:

A. Legal Authorities

This Order is issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. Environmental Protection Agency (USEPA) and Chapter 5.5, Division 7 of the California Water Code (commencing with Section 13370). It shall serve as a NPDES permit for point source discharges from this Facility to surface waters. This Order also serves as Waste Discharge Requirements (WDRs) pursuant to article 4, Chapter 4, Division 7 of the Water Code (commencing with Section 13260) and as a master reclamation permit pursuant to Section 13523.1 of Article 4, Chapter 7, Division 7 of the California Water Code. This Order, as a master reclamation permit, includes Producer/User Reclamation Requirements to regulate recycled water use for irrigation and other industrial uses. In addition, this Order includes reclamation specification for the use of recycled water produced at IRWD's Facilities at use areas located within the jurisdictional boundaries of the San Diego Regional Water Board, in the City of Lake Forest, pursuant to a Regional Water Board designation request made by IRWD pursuant to Section 13228(a) of the California Water Code. The San Diego and this Regional Water Boards agreed with IRWD's request to designate this Regional Water Board as the sole Regional Water Board for regulating the IRWD's Water Recycling Plants and this designation will become effective once the San Diego Regional Water Board amends Order 97-52 or adopts a new master reclamation permit for the South Orange County Wastewater Authority that excludes IRWD.

38. Attachment F, Applicable Plans, Policies, and Regulations, page F-17, replace Section III.B., as follows:

B. California Environmental Quality Act (CEQA)

Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of CEQA, Public Resources Code section 21000 et seq. (*County of Los Angeles v. California State Water Resources Control Board* (2006) 143 Cal.App.4th 985, mod. (Nov. 6, 2006, B184034) 50 Cal.Rptr.3d 619, 632-636.) For the plant expansion project, a mitigated negative declaration and addendum was adopted on July 2, 2003 and October 5, 2005, respectively. This action also involves the re-issuance of waste

discharge requirements for an existing facility that discharges treated wastewater to land and as such, is exempt from the provisions of California Environmental Quality Act (commencing with Section 21100) in that the activity is exempt pursuant to Title 14 of the California Code of Regulations Section 15301". MWRP has completed the capacity expansion project Environmental Impact Report, SCH 2005051174 on December 28, 2005. The EIR was certified on February 27, 2006. The CEQA documents for discharge location DP-007, Syphon Reservoir, were completed on December 10, 2009 and a negative declaration was certified on December 14, 2009. The Final Supplemental Environmental Impact Report (SEIR) No. 1 for the MWRP Phase 2 and 3 Capacity Expansion Project, Biosolids Handling Component, SCH 2011031091, was certified on October 23, 2012. The SEIR No. 1 covered the CEQA process for the Biosolids Handling and Energy Recovery Facility and includes the mitigation of potential odor sources with an adequate odor control system, monitoring for presence of endangered species and/or installation of noise control measures during nesting season, and considers measures to protect human health and safety during construction. The CEQA document, Addendum No. 4 to the Michelson Water Reclamation Plant Phase 2 and 3 Capacity Expansion Project (Final EIR), addresses the temporary emergency discharge of tertiary treated effluent through Discharge Point Storm-007 to Reach 1 of San Diego Creek and was completed on March 2018 and a negative declaration was certified by IRWD on March 26, 2018. Regional Board staff has reviewed the CEQA documents related these projects and finds that if the Discharger complies with the requirements specified in this Order, it should not have any significant impact on the water quality of the receiving water bodies.

39. Attachment F, Applicable Plans, Policies, and Regulations, page F-17, replace Section III.C.1. and Table 10, as follows:

1. **Water Quality Control Plans.** The Regional Water Board adopted an updated Water Quality Control Plan for the Santa Ana Basin (hereinafter Basin Plan) that became effective on January 24, 1995. The Basin Plan designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters addressed through the plan. On January 22, 2004, the Regional Water Board adopted Resolution No. R8-2004-0001, amending the Basin Plan to incorporate revised boundaries for groundwater subbasins, now termed "management zones", new nitrate-nitrogen and TDS objectives for the new management zones, and new nitrogen and TDS management strategies applicable to both surface and ground waters. The State Water Board and Office of Administrative Law (OAL) approved the N/TDS Amendment on September 30, 2004 and December 23, 2004, respectively. EPA approved the surface water standards components of the N/TDS Amendment on June 20, 2007. Effluent limitations for TDS and TIN in this Order are based on applicable water quality objectives, adopted and now in effect as part of the N/TDS Amendment. TDS

limits for recycled water use are based on the TDS objective for the affected groundwater management zone.

The San Diego Regional Water Board adopted the Water Quality Control Plan for their region on September 8, 1994 and subsequently approved by the State Water Resources Control Board (SWRCB) on December 13, 1994. Subsequent revisions to the San Diego Regional Water Board's Basin Plan have also been adopted by their board and approved by the SWRCB. The San Diego Regional Water Board's Basin Plan contains beneficial uses and water quality objectives. The reclamation specifications for recycled water use, within the Aliso Creek Hydrologic Sub-area, of this Order are consistent with the San Diego Regional Water Board's Basin Plan.

In addition, State Water Resources Control Board (State Water Board) Resolution No. 88-63 (Sources of Drinking Water Policy) requires that, with certain exceptions, the Regional Water Board assign the municipal and domestic water supply use to water bodies. Based on the exception criteria specified in Resolution No. 88-63, the Regional Water Board excepted Reach 1 and 2 of the San Diego Creek, Rattlesnake, Syphon, and Sand Canyon Reservoirs, and Upper and Lower Newport Bay from the municipal and domestic supply beneficial use. The Aliso Creek Hydrologic Sub-area is not exempted from the MUN designation under the San Diego Regional Water Board's Basin Plan.

As previously discussed, the Facility discharges wastewater from several discharge points. The designated beneficial uses of receiving waters affected by the discharge from the Facility are as follows:

Table 10. Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
001 to 08	Irvine Groundwater Management Zone	<u>Present or Potential:</u> Municipal and domestic supply, agricultural supply, industrial service supply, and industrial process supply.
003	Rattlesnake Reservoir	<u>Present or Potential:</u> Agricultural supply, water contact recreation, non-contact water recreation, warm freshwater habitat, and wildlife habitat Excepted from Municipal and Domestic Supply
004	Sand Canyon Reservoir	<u>Present or Potential:</u> Agricultural supply, water contact recreation, non-contact water recreation, warm freshwater habitat, and wildlife habitat Excepted from Municipal and Domestic Supply
007	Syphon Reservoir	<u>Present or Potential:</u> Agricultural supply, water contact recreation, non-contact water recreation,

Table 10. Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
		warm freshwater habitat, and wildlife habitat Excepted from Municipal and Domestic Supply
001 & 008	Aliso Creek Hydrologic Sub-area (Per San Diego Regional Board's Basin Plan)	<u>Present or Potential:</u> Municipal and domestic supply and agricultural supply.
Storm-007	San Diego Creek, Reach 1	<u>Present or Potential:</u> Water contact recreation, non-contact water recreation, warm freshwater habitat, and wildlife habitat Excepted from Municipal and Domestic Supply
Storm-008	San Joaquin Freshwater Marsh	<u>Present or Potential:</u> Water contact recreation; non-contact water recreation; warm freshwater habitat; preservation of biological habitats of special significance; rare, threatened or endangered species; and wildlife habitat Excepted from Municipal and Domestic Supply
Storm-007 & Storm-008	Newport Bay, Upper	<u>Present or Potential:</u> Water contact recreation, Non-contact water recreation, Commercial and sport fishing, Preservation of biological habitats of special significance, Wildlife habitat, Rare, threatened or endangered species, Spawning, reproduction, and development, Marine habitat, Shellfish harvesting, and Estuarine habitat Excepted from Municipal and Domestic Supply

Requirements of this Order implement the Basin Plans of this Regional Water Board and the San Diego Regional Water Board.

40. Attachment F, Applicable Plans, Policies, and Regulations, page F-21, add Section III.C.10., as follows:

10. Water Quality Control Policy for the Enclosed Bays and Estuaries of California. This policy was amended by SWRCB's Resolution No. 95-84 on November 16, 1995 and prohibits the discharge of new discharges of municipal wastewaters and industrial process waters (exclusive of cooling water discharges) to enclosed bays and estuaries, other than the San Francisco Bay-Delta system, which are not consistently treated and discharged in a manner that would enhance the quality of receiving waters above that which would occur in the absence of the discharge. In order to comply with this policy, the Discharger must offset TN and TP loads (lbs/rainfall year) discharged under emergency conditions from the MWRP to Newport Bay at a minimum ratio 1.5:1 in accordance with an approved TN and TP offset plan. The Discharger has committed to minimize the TN and TP loads discharged to Reach 1 of San Diego Creek by implementing enhanced nutrient removal with operational objectives of producing recycled water with a monthly average total nitrogen and total phosphorus concentrations of 10

mg/L and 0.3 mg/L respectively during diversions. The MWRP will be operated in such a manner that the Discharger will meet these TN and TP effluent concentrations as an emergency discharge facility performance goal.

41. Attachment F, Applicable Plans, Policies, and Regulations, page F-22, replace the second paragraph of Nutrient TMDL of Section III.E., as follows:

Like prior Order No. R8-2007-0003, this Order implements relevant requirements of the nutrient TMDL. The nutrient TMDL excludes total nitrogen (TN) loads from sources during storm events that result in flows of 50 mean daily cubic feet per second or more in San Diego Creek at Campus Drive, however, total phosphorous (TP) loads are not excluded under any flow conditions. To implement the nutrient TMDL, this Order includes a TN effluent limit of 1 mg/L for stormwater, recycled water and tertiary treated effluent discharges to San Diego Creek, which is tributary to Newport Bay. TN and TP loads associated with the emergency discharge of stormwater or tertiary treated effluent (from MWRP) and recycled water overflows (from Sand Canyon Reservoir) to San Diego Creek, which cannot be excluded and are in excess of 1 mg/L for TN, must be offset at a minimum ratio of 1.5:1. This Order includes Discharge Prohibition III.F.4. that addresses this condition by requiring an approved nutrient offset program for recycled water overflows from Sand Canyon Reservoir to San Diego Creek in excess of 1 mg/L of TN. Also, during wet weather conditions, storm water runoff from the Sand Canyon Reservoir's watershed occasionally causes overflows³ from the reservoir into San Diego Creek. The Discharger has implemented best management practices to minimize overflows from Sand Canyon Reservoir during the rainy season. During the term of Order No. R8-2007-0003, Sand Canyon Reservoir has overflowed twice in 2010 and 2011 during the rainy season. Recent monitoring data indicate that the nutrient TMDL goals for 2012 have already been met. The Discharger has complied and proposes to continue to comply with these requirements by implementing an approved TN and TP offset program as required by this Order.

The TN and TP offset program will consist of computing and reporting the TN and TP loads that the Discharger removes from the San Diego Creek and Newport Bay Watershed, during the rainfall year (October through September), through the operations of IRWD's natural treatment system (NTS) wetlands, including the San Joaquin Freshwater Marsh and urban runoff diversions. TN and TP mass loads discharged to San Diego Creek must be offset with TN and TP mass removal credits accumulated within a three-rainfall year period that includes the rainfall year in which the discharge occurred. IRWD's treatment wetlands in the San Joaquin Marsh have removed nearly one million pounds of nitrogen since beginning full operation in 1999. More recent annual nitrogen removals have decrease due to declining flow rates and nitrogen concentrations in San Diego Creek, but the wetlands still removed 28,700 lbs of nitrogen in 2016-17. IRWD has documented up to 50,000 lbs of additional nitrogen removals for 2016-17 from its other NTS

facilities in the San Diego Creek watershed. Unlike nitrogen, the IRWD wetlands in the San Joaquin Marsh do not provide reliable phosphorus removal. However, IRWD's other NTS facilities removed a total of 1,181 lbs of phosphorus in 2016-17 and has removed a cumulative total of 18,914 lbs of phosphorus since beginning full operation in 1999. The 1.5:1 offset ratio is required to comply with the prohibition on new treated wastewater discharges included in the Water Quality Control Policy for the Enclosed Bays and Estuaries of California, which requires that new treated wastewater discharges enhance the water quality of the receiving water body (Newport Bay).

42. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-25, replace the title of Section IV.C., as follows:

C. Water Quality-Based Effluent Limitations (WQBELs) for DPs 003, 004, 007 & Storm-007 (Tertiary treated effluent)

43. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-26 and F-27, replace the paragraphs for Total Chlorine Residual, TDS/Ammonia-Nitrogen and Table 12 of Section IV.C.2.a., as follows:

Total Chlorine Residual (TCR): Recycled water from the MWRP is distributed to the Sand Canyon, Syphon, and Rattlesnake Reservoirs for storage. Super-chlorination of the recycled water is necessary to prevent fouling of the recycled water distribution lines. As such, discharges of recycled water into the reservoirs contain concentrations of residual chlorine that may be toxic to aquatic organisms that may be present in the reservoirs. However, the residual chlorine present in the discharges dissipates within the reservoirs such that, currently, the reservoirs support aquatic life, including fish. The discharge of chlorinated recycled water to the reservoirs does not compromise the beneficial uses of the reservoirs, which include warm water aquatic habitat. The discharge of chlorinated recycled water to the reservoirs is necessary to accommodate wastewater reclamation and water conservation. It is in the public interest to accommodate these activities.

This Order specifies an effluent limitation for TCR to protect aquatic life beneficial uses. Compliance with this limitation, at Rattlesnake and Sand Canyon Reservoirs, is to be achieved outside a zone of initial dilution (about 45 feet away from the outfall at each reservoir), wherein dissipation of the chlorine occurs. In the case of Syphon Reservoir, due its smaller volume with respect to the other two reservoirs, compliance is to be achieved at the point of discharge to the reservoir (DP 007). For tertiary treated effluent discharges through Storm-007, compliance with the TCR effluent limit shall be achieved at the discharge point to San Diego Creek.

TDS/Ammonia-Nitrogen: TDS/Ammonia-Nitrogen limitations are specified in the Order for discharges to surface waters, including Sand Canyon, Syphon, and

Rattlesnake Reservoirs and Reach 1 of San Diego Creek. The proposed TDS limit for the reservoirs is based on the prior effluent limit in Order No. R8-2007-0003 and for emergency tertiary treated effluent discharges to Reach 1 of San Diego Creek is 1,500 mg/L. The proposed total ammonia-nitrogen monthly average limit was updated and calculated using the equations included in Table 4-4 of Chapter 4 of the Basin Plan. The numeric value of the proposed total ammonia-nitrogen limit is less than the limit included in Order No. 2007-0003 and details regarding the calculation of the proposed limit are included in Attachment O of this Order. For short-duration emergency discharge of tertiary treated effluent to Reach 1 of San Diego Creek the chronic criteria derived effluent limit applies to protect the WARM beneficial use of the reservoirs and as a facility performance criterion. The TDS/ Total Ammonia-Nitrogen limits are shown in the table below.

Table 12. TDS/Ammonia-Nitrogen Limits

Discharge Point	TDS, mg/L	Total Ammonia-Nitrogen, mg/L
003, 004 & 007	720	0.75
Storm-007	1,500	0.75

44. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-30, replace title of Table 14., as follows:

Table 14. RPA Summary for DPs 003, 004, 007 & Storm-007

45. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-31, replace title of Section IV.D., as follows:

D. Best Professional Judgment-Based Effluent Specifications for DPs 003, 004, 007 & Storm-007

46. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-32, replace title of Section IV.E., as follows:

E. Summary of Final Effluent Limitations for DPs 003, 004, 007 & Storm-007

47. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-32, replace Section IV.E.2., as follows:

2. Satisfaction of Antidegradation Policy

Discharges in conformance with the requirements of this Order will not result in a lowering of water quality and therefore conform to antidegradation requirements specified in Resolution No. 68-16, which incorporates the federal antidegradation policy at 40 CFR 131.12. IRWD has significantly improved the quality of domestic water supply to its service area through construction and operation of the Irvine Desalter Project (see section II.A.5.),

reduced reliance on high TDS imported water and use of low TDS groundwater. During dry weather, all the treated effluent from these Facilities is and will continue to be recycled or in the case of LAWRP, the treated effluent will be discharged to the Pacific Ocean. During periods of low demand for recycled water at MWRP, wastewater that is not needed for recycling is diverted to OCSD facilities for treatment and ocean disposal or to the Orange County Water District's Groundwater Replenishment System. Implementation of the Groundwater Replenishment Project, a joint effort of the OCSD (which includes the Discharger) and the Orange County Water District has created and will create additional demand for recycled water. The Discharger will comply with the prohibition for new treated wastewater discharges included in the Water Quality Control Policy for the Enclosed Bays and Estuaries of California by offsetting TN and TP loads (lbs/rainfall year) discharged under temporary emergency conditions from the MWRP to Reach 1 of San Diego Creek, which is tributary to Newport Bay, at a minimum ratio of 1.5:1 in accordance with an approved TN and TP offset plan. The Discharger has committed to minimize the TN and TP loads that could potentially be discharged to Newport Bay by implementing enhanced nutrient removal with operational objectives of producing recycled water with a monthly average total nitrogen and total phosphorus concentrations of 10 mg/L and 0.3 mg/L respectively during diversions. The MWRP will be operated in such a manner that the Discharger will meet these TN and TP effluent concentrations as an emergency discharge facility performance goal. The Water Quality Control Policy for the Enclosed Bays and Estuaries of California requires that new treated wastewater discharges enhance the water quality of the receiving water body (Newport Bay) if authorized. The limitations in this Order are established at levels that assure no degradation of groundwater will occur as the result of wastewater recycling. Therefore, this discharge is consistent with federal and state antidegradation policies.

48. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-33, replace title of Section IV.E.4. and Table 16., as follows:

4. Summary of Final Effluent Limitations for DPs 003, 004, 007 & Storm-007:

Table 16. Summary of Water Quality-based Final Effluent Limitations for Discharge Points 003, 004, 007 & Storm-007

Parameter	Units	Effluent Limitations					Basis
		Average Monthly or as noted herein	Average Weekly	Max Daily	Instantaneous Minimum	Instantaneous Maximum	
BOD ₅	mg/L	20	30	--	--	--	BPJ,PO
Total Suspended	mg/L	20	30	--	--	--	BPJ,PO

Table 16. Summary of Water Quality-based Final Effluent Limitations for Discharge Points 003, 004, 007 & Storm-007

Parameter	Units	Effluent Limitations					Basis
		Average Monthly or as noted herein	Average Weekly	Max Daily	Instantaneous Minimum	Instantaneous Maximum	
Solids							
Total Residual Chlorine	mg/L	--	--	--	--	0.1	PO, BP
Ammonia Nitrogen	mg/L	0.75					BP
TDS	mg/L	720/1,500 (12-M avg)	--	--	--	--	PO, BP
pH	Std. unit	--	--	--	6.5	8.5	PO, BP
Coliform	MPN	--	2.2 Median of last 7 days	--	--	--	PO, Title 22
Dichlorobromo methane	µg/L	46		71			CTR, SIP

Notes: PO = Previous Order, R8-2007-0003; BP= Basin Plan. The 1,500 mg/L TDS limit applies only at Storm-007.

49. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, Reclamation Specifications, page F-34, replace Section IV.H.3. and add Table 17, as follows:

3. Summary of Effluent Limitations at DPs 001, 002 and 008: The TDS limitation of 910 mg/L for recycled water reuse on sites overlying the Irvine Groundwater Management Zone is based on the TDS water quality objective in Table 5-3 of the Basin Plan. Also, the TDS limitation of 1,000 mg/L for recycled water reuse on sites overlying the Aliso Creek Hydrologic Sub-area is based on the TDS water quality objective included in the Basin Plan of the San Diego Regional Water Board. The effluent limits for biochemical oxygen demand and total suspended solids are based on best professional judgment (Section IV.D of Attachment F of this Order) and pH is based on water quality objectives included in the Basin Plans of this Regional Water Board and the San Diego Regional Water Board. All other recycled water effluent limits included in Table 17 below are based on the water quality objectives that apply to the Aliso Creek Hydrologic Sub-area; which are included in the Basin Plan of the San Diego Regional Water Board.

Table 17 Recycled Water Effluent Limitations at DPs 001, 002 & 008

Parameter	Units	Effluent Limitations		
		Average Monthly	Average Weekly	12-month running average
Biochemical Oxygen Demand	mg/L	20	30	--

Table 17 Recycled Water Effluent Limitations at DPs 001, 002 & 008

Parameter	Units	Effluent Limitations		
		Average Monthly	Average Weekly	12-month running average
5-day @ 20°C				
Total Suspended Solids	mg/L	20	30	- -
pH	standard units	6 (instantaneous minimum)	9 (instantaneous maximum)	- -
Chloride	mg/L			375
Sulfate	mg/L			375
% Sodium	%			60
Nitrate	mg/L			45
Iron	mg/L			0.3
Manganese	mg/L			0.05
MBAS	mg/L			0.5
Boron	mg/L			0.75
Aluminum	mg/L			1.0
Arsenic	mg/L			0.05
Antimony	mg/L			0.006
Barium	mg/L			1.0
Beryllium	mg/L			0.004
Cadmium	mg/L			0.005
Cyanide	mg/L			0.2
Mercury	mg/L			0.002
Nickel	mg/L			0.1
Selenium	mg/L			0.05
Thallium	mg/L			0.002

Note: The 12-month running average effluent limitations on this table apply only to recycled water use that overlies the Aliso Creek Hydrologic Sub-area within the San Diego Regional Water Board jurisdictional boundary.

50. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-35, renumber Table 17 as Table 18, as follows:

Table 18. Stormwater Discharge Limits at Storm-007 & Storm-008

51. Attachment F, Rationale for Receiving Water Limitations, page F-36, replace Section V.A.2., as follows:
- The previous Order R8-2007-0003 allowed discharges from Sand Canyon Reservoir under certain storm conditions. This Order also allows discharges under extreme storm conditions. The nutrient TMDL excludes nitrogen loads from sources during storm events that result in flows of 50 mean daily cubic feet per second or more in San Diego Creek at Campus Drive. This Order includes Discharge Prohibition III.F.4, which addresses this condition by requiring an approved nutrient offset program for recycled water discharges

exceeding 1 mg/L of total nitrogen. In addition, to address the potential of a temporary emergency discharge of tertiary treated effluent from MWRP to Reach 1 of San Diego Creek, which is tributary to Newport Bay, the Order requires the Discharger to implement an approved TN and TP offset plan that can mitigate TN and TP loads discharged to Newport Bay at a ratio of 1.5:1 and would minimize the TN and TP loads discharged by operating the MWRP in accordance with a facility performance goal. This Order specifies that this type of episodic emergency discharge is authorized under emergency conditions resulting from extreme rainfall events and/or season, as last resort disposal and after the Discharger has exhausted its recycled water storage capacity, has maximized the disposal of recycled water through OCSD's ocean outfall and it is not feasible to dispose of sewage through OCSD's sewage treatment plants (amongst other conditions). This Order recognizes the efforts of the Discharger to expand its recycled water storage capacity and work to evaluate the possibility of a permanent emergency discharge point to Reach 1 of the Santa Ana River, which would not have the current regulatory constraints.

52. Attachment F, Rationale for Receiving Water Limitations, page F-36, replace Section V.B., as follows:

C. Groundwater

The receiving groundwater limitations in the proposed Order are based upon the water quality objectives contained in the Basin Plans of this Regional Water Board and the San Diego Regional Water Board.

53. Attachment F, Rationale for Provisions, page F-40, add Section VII.B.2.d., as follows:

- d. Discharger is required to submit for approval by the Executive Officer a TN and TP Offset plan that details the manner in which all pertinent information will be gathered to document the removal credits of TN and TP loads in the NTS, including the San Joaquin Freshwater Marsh and urban runoff diversions, managed by IRWD and the reporting of the data (See Section VI.C.2.e. of this Order).

54. Attachment F, Rationale for Provisions, page F-41, add Section VII.B.6.a., as follows:

- a. **Emergency Conditions for Discharge of Tertiary Treated Effluent to San Diego Creek:** The Discharger is authorized to discharge tertiary treated effluent to Reach 1 of San Diego Creek, through discharge point Storm-007, on a temporary emergency basis, provided that the Discharger meets several conditions (See Section VI.C.6.b. of this Order).

56. Attachments, add Attachment Q - Santa Ana Regional Board's Designation Agreement Letter, pages Q-1 through Q-3, which is attached to this Order.
57. These amendments shall become effective upon the adoption of this Order.
58. All other conditions and requirements of Order No. R8-2015-0024, including Attachments, shall remain unchanged.

I, Hope A. Smythe, Executive Officer, do hereby certify that the forgoing is a full, true, and correct copy of an order adopted by the California Regional Water Quality Control Board, Santa Ana Region, on September 7, 2018.


Hope A. Smythe
Executive Officer

California Regional Water Quality Control Board
Santa Ana Region

September 7, 2018

STAFF REPORT

ITEM: *6

SUBJECT: Amendment of Order No. R8-2015-0024, NPDES No. CA8000326, Waste Discharge Requirements and Master Reclamation Permit for Irvine Ranch Water District, Water Recycling Plants, Orange County – Order No. R8-2018-0070

DISCUSSION:

On June 19, 2015, the Santa Ana Regional Board adopted Order No. R8-2015-0024, NPDES No. CA8000326, prescribing waste discharge requirements and master reclamation permit for the Irvine Ranch Water District's (hereinafter Discharger or IRWD's) water recycling plants, surface waters discharges and recycled water use within the Irvine Groundwater Management Zone.

The Discharger owns and operates the Michelson and Los Alisos Water Recycling Plants (WRPs) and a recycled water distribution system that includes lined and unlined reservoirs, pump stations, and pipelines, among other features, that distribute recycled water within the boundaries of the Santa Ana Regional Board and the San Diego Regional Water Quality Control Board (San Diego Regional Board). However, IRWD's water recycling plants, reservoirs and most of the recycled water use areas are located within the boundaries of the Santa Ana Regional Board and a small fraction of recycled water use areas are located within the San Diego Regional Board.

Currently, the San Diego Regional Board regulates the use of recycled water, produced at the Discharger's WRPs, within the Aliso Creek Hydrologic Sub-area under Order 97-52, *Waste Discharge and Water Recycling Requirements for the Production and Purveyance of Recycled Water by Member Agencies of the South Orange County Reclamation Authority*, which is a regional water recycling permit.

In a letter entitled, "Request for Santa Ana Regional Water Quality Control Board to Regulate Waste Discharge Requirements from Irvine Ranch Water District Water Recycling Plants," dated March 30, 2016, the Discharger requested to this Regional Board and the San Diego Regional Board the designation of the Santa Ana Regional Board as the sole Regional Board for the regulation of the treatment, distribution and use of recycled water produced at the Discharger's water recycling plants. The Discharger made the regional board designation request pursuant to Section 13228(a) of the California Water Code, which states the following:

(a) Concerning any matter that may be submitted to a regional board by a person or entity that is subject to regulation by more than one Regional Board, the

person or entity may submit the matter to one of those regional boards if both of the following requirements are met:

(1) The person or entity submits a written request to all affected regional boards that one Regional Board be designated to regulate the matter.

(2) All affected Regional Boards agree in writing to the designation. Unless the Board of any affected Regional Board denies the request, the Executive Officer of a Regional Board may grant a request submitted pursuant to paragraph (1) on behalf of that Board.

Also, the Discharger's March 30, 2016 letter included supporting information such as Exhibits "A" and "B". Exhibit "A" consists of a map that shows the location of IRWD's water recycling plants, recycled water reservoirs and distribution pipelines within the boundaries of the San Diego and Santa Ana Regional Boards. Exhibit "B" contains a comparison between the annual averages of recycled water monitoring data for the last ten years for both IRWD's water recycling plants and the 12-month average effluent limits for recycled water discharges specified under San Diego Regional Board's Order 97-52. The recycled water monitoring data indicates that IRWD has complied with all effluent limits for the last ten years.

The San Diego and the Santa Ana Regional Boards agreed with IRWD's request to designate the Santa Ana Regional Board as the sole Regional Water Board for regulating the IRWD's Water Recycling Plants. In a letter dated July 22, 2016, which was addressed to IRWD, the San Diego Regional Board memorialized this agreement and included pertinent background and supporting information and conditions for the designation responsibility of the Santa Ana Regional Board, which becomes Attachment "P" of Order No R8-2015-0024. The Santa Ana Regional Board ratified this agreement in a letter issued to IRWD dated September 19, 2016, which becomes Attachment "Q" of Order No. R8-2015-0024. This designation will become effective once the San Diego Regional Water Board amends Order 97-52 or adopts a new master reclamation permit for the South Orange County Wastewater Authority to exclude IRWD.

On July 18, 2017 the Discharger submitted a Report of Waste Discharge (Form 200) with the purpose of requesting authorization to discharge up to 28 MGD of tertiary treated and disinfected effluent, under emergency conditions, to Reach 1 of San Diego Creek through Discharge Point Storm-007 (Storm-007). Through June 6, 2018, the Discharger submitted several documents in support of the proposed discharge and in response to information request from this Regional Water Board staff to demonstrate that the proposed discharge does not impact San Diego Creek and Newport Bay and complies with the conditions of the Nutrient Total Maximum Daily Load (TMDL) and other TMDLs applicable to the San Diego Creek/Newport Bay watershed and the Water Quality Control Policy for the Enclosed Bays and Estuaries of California. The proposed discharge to Reach 1 of San Diego Creek would occur under emergency conditions where an extreme rainfall event and/or season causes IRWD to run out of recycled water storage, demand for recycled water to be very low, and may not be feasible to

send raw sewage, usually treated at Michelson WRP, to Orange County Sanitation District's (OCSD's) wastewater treatment plants (WWTPs) for treatment and disposal.

The nutrient TMDL excludes total nitrogen (TN) loads that are discharged to the San Diego Creek/Newport Bay watershed when flow in San Diego Creek is equal or greater than 50 cubic-feet-per-second (cfs) at Campus Drive, however, it does not exclude total phosphorous (TP) loads discharged under any flow conditions in San Diego Creek. The Water Quality Control Policy for the Enclosed Bays and Estuaries of California, as amended by SWRCB's Resolution No. 95-84 on November 16, 1995, prohibits the discharge of new discharges of municipal wastewaters and industrial process waters (exclusive of cooling water discharges) to enclosed bays and estuaries, other than the San Francisco Bay-Delta system, which are not consistently treated and discharged in a manner that would enhance the quality of receiving waters above that which would occur in the absence of the discharge. To comply with this policy and the nutrient TMDL, the Discharger must offset TN and TP loads (lbs/rainfall year) discharged under emergency conditions from the MWRP to Newport Bay at a minimum ratio 1.5:1 in accordance with an approved TN and TP offset plan. The TN and TP offset program will consist of computing and reporting the TN and TP loads that the Discharger removes from the San Diego Creek and Newport Bay Watershed through IRWD's operations of NTS wetlands, including the San Joaquin Marsh, and urban runoff diversions. IRWD must document the TN and TP offsets accumulated over a 3-rainfall year period that includes the rainfall year (October through September) when the effluent discharge occurred. In addition, the Discharger has committed to minimize the TN and TP loads discharged to Reach 1 of San Diego Creek by implementing enhanced nutrient removal with operational objectives of producing average recycled water total nitrogen and total phosphorus concentrations of 10 mg/L and 0.3 mg/L respectively during diversions. The MWRP will be operated in such a manner that the Discharger will meet these TN and TP effluent concentrations as an emergency discharge facility performance goal.

Normally, the Discharger treats and produces at the Michelson WRP an average monthly flow rate of 20 MGD of tertiary-2.2 quality recycled water that is delivered to use sites and/or stored and during the rainy season, when demand is low, the recycled water can also be supplied to the Orange County Water District's (OCWD's) Green Acres Project (GAP). Under very low demand circumstances and when storage reservoirs are at almost maximum operating capacity, the Discharger currently has the ability to re-route most of the raw sewage flow that is treated at the Michelson WRP to OCSD's wastewater treatment plants (WWTPs) for treatment/disposal and/or sends recycled water to OCSD Plant No. 2 for disposal through their Ocean outfall.

However, during a large rain storm event that occurred during February 2017, OCSD's WWTPs experienced very high influent peak flows that significantly stressed the hydraulic capacity of their headworks; which could have been alleviated by reducing the raw sewage influent flow received from controllable sources such as IRWD's sanitary sewer system. As a result, as another large rain storm event had been forecasted for the Orange County area during February 2017 and to avoid a similar headworks' hydraulic overload episode OCSD staff requested that IRWD temporarily cease sending raw sewage flows, normally treated at the Michelson WRP, to OCSD's WWTPs. At the

time, IRWD was running short of recycled water storage capacity, recycled water demand was very low, and could only discharge less than half of their recycled water flow through OCSD's Ocean outfall. Although the magnitude of the forecasted rainstorm event did not materialize and IRWD resumed their discharge of sewage to OCSD's WWTPs after the rain storm event ceased, the possibility exists that during an extended and intense rainy season this type of episodes might reoccur. Additionally, when the Discharger's solids handling and biosolids facility, currently under construction, comes online, IRWD's ability to divert sewage away from Michelson WRP to OCSD WWTPs will be limited. Therefore, it is warranted to allow IRWD to discharge, during brief periods and under the emergency conditions explained above, to Reach 1 of San Diego Creek to avoid an unauthorized release of raw sewage from the IRWD's and/or OCSD's sanitary sewer system and/or uncontrolled discharge of recycled water from IRWD's recycled water system (amongst other potential consequences). IRWD is pursuing the expansion of its recycled water storage capacity at Syphon Reservoir to almost 10 times the current reservoir capacity. IRWD is working towards developing an emergency discharge point to Reach 1 of the Santa Ana River for tertiary treated effluent. Before IRWD discharges to Reach 1 of San Diego Creek, the Discharger must meet several conditions and notify this Regional Water Board in advance regarding their intent to discharge

It is appropriate to amend Order No. R8-2015-0024 to modify the reclamation specifications that apply to Discharge Point 008 (DP 008) and incorporate discharge specifications that are included in San Diego Regional Board's Order 97-52 for discharges of recycled water into use areas located within the jurisdictional boundaries of the San Diego Regional Board and regulate them under Order No. R8-2015-0024. This amendment adds Attachments "P" and "Q" to Order No. R8-2015-0024. In addition, it is appropriate to amend Order No. R8-2015-0024 to modify and/or add pertinent requirements and provisions to regulate the discharge of tertiary treated effluent through Discharge Point Storm-007 to Reach 1 of San Diego Creek under extreme emergency conditions.

The following shows the proposed changes to Order No. R8-2015-0024. Additions are bold and highlighted. Deletions are stricken-out.

1. Order No. R8-2015-0024, page 1, replace Table 2., as follows:

Table 2. Discharge Locations

Discharge Point	Effluent Description	Discharge Point Latitude	Discharge Point Longitude	Receiving Water
001	Recycled water from MWRP to distribution system	33°39'50"N	117°50'17"W	Irvine GMZ
002	Recycled water from MWRP to San Joaquin Reservoir	33°37'12"N	117°50'39"W	San Joaquin Reservoir & Irvine GMZ
003	Tertiary treated effluent from MWRP	33°43'37"N	117°44'27"W	Rattlesnake Reservoir & Irvine GMZ
004	Tertiary treated effluent from MWRP	33°38'55"N	117°47'47"W	Sand Canyon Reservoir & Irvine GMZ

Table 2. Discharge Locations

Discharge Point	Effluent Description	Discharge Point Latitude	Discharge Point Longitude	Receiving Water
007	Tertiary treated effluent from MWRP	33°42'36"N	117°43'56"W	Syphon Reservoir & Irvine GMZ
008	Recycled water from LAWRP and/or MWRP to distribution system	33°38'16"N	117°42'50"W	Irvine GMZ & Aliso Creek Hydrologic Sub-area (Per San Diego Regional Board's Basin Plan)
Storm-007	Pumped stormwater or tertiary treated effluent (emergency) from MWRP	33°39'47"N	117°50'16"W	San Diego Creek Reach 1 tributary to Upper Newport Bay
Storm-008	Pumped stormwater (emergency) from MWRP	33°39'51"N	117°50'27"W	San Joaquin Freshwater Marsh tributary to Upper Newport Bay

2. Order No. R8-2015-0024, List of Attachments, page 4, replace the List of Attachments, as follows:

Attachment A – Definitions	A-1
Attachment B – Map.....	B-1
Attachment C – Flow Schematic	C-1
Attachment D – Standard Provisions.....	D-1
Attachment E – Monitoring and Reporting Program	E-1
Attachment F – Fact Sheet.....	F-1
Attachment G – EPA PPL Table	G-1
Attachment H – ML Table.....	H-1
Attachment I – Triggers for Monitoring Priority Pollutants	I-1
Attachment J – Stormwater Pollution Prevention Plan	J-1
Attachment K – Stormwater Monitoring and Reporting Requirements	K-1
Attachment L – Acceptance Conditions for UV Disinfection System	L-1
Attachment M – Reasonable Potential Analysis	M-1
Attachment N – Water Quality Based Effluent Limits Calculation.....	N-1
Attachment O – Total Ammonia-Nitrogen Limit Calculation	O-1
Attachment P – San Diego Regional Board's Designation Agreement Letter	P-1
Attachment Q – Santa Ana Regional Board's Designation Agreement Letter	Q-1

3. Order No. R8-2015-0024, Findings, page 6, add paragraph 3. And 4. to Section II.A., as follows:

- 3. The San Diego Regional Water Quality Control Board (San Diego Regional Water Board) regulates the Discharger's production and distribution of recycled water and use areas, within its region's Aliso Creek Hydrologic Sub-area, under Order 97-52, "Waste Discharge and Water Recycling Requirements for the Production and Purveyance of Recycled Water by Member Agencies of the South Orange County Reclamation Authority," which is a regional water recycling permit. In a**

letter dated March 30, 2016, the Discharger requested to this Regional Water Board and the San Diego Regional Water Board the designation of this Regional Water Board as the sole Regional Water Board for the regulation of the treatment, distribution and use areas of recycled water produced at the Discharger's water recycling plants. Including the distribution of recycled water to use areas located within the boundaries of the San Diego Regional Water Board in the City of Lake Forest. The Discharger made the regional water board designation request pursuant to Section 13228(a) of the California Water Code. The San Diego and this Regional Water Boards agreed with IRWD's request to designate this Regional Water Board as the sole Regional Water Board for regulating the IRWD's Water Recycling Plants. The San Diego and this Regional Water Boards memorialized the designation agreement in separate letters addressed to the Discharger, which were dated July 22, 2016 and September 19, 2016 respectively. Copies of these designation agreement letters are included in this Order as Attachment P and Q.

4. On July 18, 2017 the Discharger submitted a Report of Waste Discharge requesting authorization to temporarily discharge tertiary treated and disinfected effluent, under emergency conditions and as last resort for disposal, to Reach 1 of San Diego Creek through Discharge Point Storm-007. The Discharger submitted several documents in support of the propose discharge and in response to information request from Regional Water Board staff to demonstrate that the proposed discharge complies with The Water Quality Control Policy for the Enclosed Bays and Estuaries of California, as amended by SWRCB's Resolution No. 95-84 on November 16, 1995, does not impact Reach 1 of San Diego Creek and Newport Bay and complies with the conditions of the nutrient Total Maximum Daily Load (TMDL) and other TMDLs (fecal coliform, sediment and toxic pollutants) established for the San Diego Creek/Newport Bay watershed. Regional Water Board staff requested and received additional information from the Discharger until June 6, 2018.

4. Order No. R8-2015-0024, Findings, page 6, replace Section II.B., as follows:

B. Facility Description. The Discharger owns and operates the MWRP and its wastewater treatment systems consist of preliminary, primary, secondary, and tertiary treatment. Recycled water is directly delivered from Discharge Point 001 (DP-001) to customers for irrigation and other recycled water reuse, except for indirect or direct potable water reuse. Recycled water from MWRP is discharged through DP-003, DP-004 and DP-007 to Rattlesnake Reservoir, Sand Canyon Reservoir, and Syphon Reservoir respectively, for storage and subsequent distribution. These three reservoirs are considered waters of the United States. **Under emergency conditions, the Discharger may discharge stormwater or tertiary treated effluent through Discharge Point Storm-007 to Reach 1 of San Diego Creek and stormwater, through**

Discharge Point Storm-008, to San Joaquin Freshwater Marsh; which are tributary to Newport Bay. The recycled water use areas overlie the Irvine Groundwater Management Zone **and the San Diego Regional Water Board's Aliso Creek Hydrologic Sub-area.** Also, recycled water can be routed to the Orange County Sanitation District (OCSD) for disposal through their ocean outfall and/or Orange County Water District's (OCWD) Green Acres Project's (GAP's) recycled water distribution system. Both OCSD's ocean outfall discharge and OCWD's GAP are regulated by this Regional Water Board under separate permits.

Because of high groundwater elevation at the MWRP site, dewatering of the shallow groundwater zone is necessary to protect in-ground facilities. The area is dewatered through a network of shallow zone wells with two separate discharge points. These discharge points discharge either into Reach 1 of San Diego Creek or into the San Joaquin Freshwater Marsh. Reach 1 of San Diego Creek and San Joaquin Freshwater Marsh are tributaries to San Diego Creek and waters of the U.S. Order No. R8-2007-0003 also regulated the discharges from these dewatering wells; however, this Order does not regulate the groundwater dewatering discharges from the MWRP (see Attachment F of this Order for details).

Also, the Discharger owns and operates the LAWRP and its wastewater treatment processes include preliminary, secondary, and tertiary treatment. **Furthermore, the Discharger distributes from LAWRP Tertiary-2.2 recycled water, which is produced at the Facilities, is delivered from through DP-008 to use areas customers located within the boundaries of this Regional Water Board and the San Diego Regional Water Board, in the City of Lake Forest,** for irrigation and other recycled water reuse, except groundwater recharge. Alternatively, the LAWRP may discharge undisinfected secondary treated effluent to the Pacific Ocean through the Aliso Creek ocean outfall operated by the SOCWA. The discharge to the Pacific Ocean is regulated by the San Diego Regional Water Board through Order No. R9-2012-0013 issued to SOCWA.

5. Order No. R8-2015-0024, Findings, page 7, replace Section II.C., as follows:

C. Legal Authorities. This Order is issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. Environmental Protection Agency (USEPA) and Chapter 5.5, Division 7 of the California Water Code (commencing with Section 13370). It shall serve as a NPDES permit for point source discharges from the MWRP to surface waters. This Order also serves as Waste Discharge Requirements (WDRs) pursuant to Article 4, Chapter 4, Division 7 of the California Water Code (commencing with section 13260) and as a master reclamation permit pursuant to Section 13523.1 of Article 4, Chapter 7, Division 7 of the California Water Code. **In addition, this Order includes reclamation specification for the use of recycled water produced at IRWD's Facilities at use areas located within**

the jurisdictional boundaries of the San Diego Regional Water Board, in the City of Lake Forest, in accordance with a Regional Water Board designation request made by IRWD pursuant to Section 13228(a) of the California Water Code. The San Diego and this Regional Water Boards agreed with IRWD's request to designate this Regional Water Board as the sole Regional Water Board for regulating the IRWD's Water Recycling Plants and this designation will become effective once the San Diego Regional Water Board amends Order 97-52 or adopts a new master reclamation permit for the South Orange County Wastewater Authority that excludes IRWD.

6. Order No. R8-2015-0024, Findings, page 7, replace Section II.E., as follows:

E. California Environmental Quality Act (CEQA). Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of CEQA, Public Resources Code section 21000 et seq. (County of Los Angeles v. California State Water Resources Control Board (2006) 143 Cal.App.4th 985, mod. (Nov. 6, 2006, B184034) 50 Cal.Rptr.3d 619, 632-636). This action also involves the re-issuance of waste discharge requirements for existing Facilities that discharge treated wastewater to land and as such, is exempt from the provisions of the California Environmental Quality Act (commencing with Section 21100) in that the activity is exempt pursuant to Title 14 of the California Code of Regulations Section 15301. The Discharger completed the MWRP capacity expansion project Environmental Impact report, SCH 2005051174 on December 28, 2005. The EIR was certified on February 27, 2006. The CEQA documents for discharge location DP-007, Syphon Reservoir, were completed on December 10, 2009 and a negative declaration was certified on December 14, 2009. The Final Supplemental Environmental Impact Report (SEIR) No. 1 for the MWRP Phase 2 and 3 Capacity Expansion Project, Biosolids Handling Component, SCH 2011031091, was certified on October 23, 2012. The SEIR No. 1 covered the CEQA process for the Biosolids Handling and Energy Recovery Facility and includes the mitigation of potential odor sources with an adequate odor control system, monitoring for presence of endangered species and/or installation of noise control measures during nesting season, and considers measures to protect human health and safety during construction. **The CEQA document, Addendum No. 4 to the Michelson Water Reclamation Plant Phase 2 and 3 Capacity Expansion Project (Final EIR), addresses the temporary emergency discharge of tertiary treated effluent through Discharge Point Storm-007 to Reach 1 of San Diego Creek and was completed on March 2018 and a negative declaration was certified by IRWD on March 26, 2018.** Regional Board staff has reviewed the above-referenced CEQA documents and finds that if the Discharger complies with the requirements specified in this Order, the discharges regulated under this Order should not have any significant impact on the water quality of the receiving waters.

7. Order No. R8-2015-0024, Findings, page 8 and 9, replace Section II.H. and Table 5., as follows:

H. Water Quality Control Plans. The Regional Water Board adopted a revised Water Quality Control Plan for the Santa Ana Region (hereinafter Basin Plan) that became effective on January 24, 1995. The Basin Plan designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters in the Santa Ana Region addressed through the plan. More recently, the Basin Plan was amended significantly to incorporate revised boundaries for groundwater subbasins, now termed “management zones”, new nitrate-nitrogen and TDS objectives for the new management zones, and new nitrogen and TDS management strategies applicable to both surface and ground waters.

This Basin Plan Amendment was adopted by the Regional Water Board on January 22, 2004. The State Water Resources Control Board (State Water Board) and Office of Administrative Law (OAL) approved the Amendment on September 30, 2004 and December 23, 2004, respectively. EPA approved the surface water standards components of the N/TDS Amendment on June 20, 2007. **The San Diego Regional Water Board adopted the Water Quality Control Plan for their region on September 8, 1994 and subsequently approved by the State Water Resources Control Board (SWRCB) on December 13, 1994. Subsequent revisions to the San Diego Regional Water Board’s Basin Plan have also been adopted by their board and approved by the SWRCB. The San Diego Regional Water Board’s Basin Plan contains beneficial uses and water quality objectives. The reclamation specifications for recycled water use, within the Aliso Creek Hydrologic Sub-area, of this Order are consistent with the San Diego Regional Water Board’s Basin Plan.**

In addition, the Basin Plan implements State Water Resources Control Board (State Water Board) Resolution No. 88-63, which established State policy that all waters should be considered suitable or potentially suitable for municipal or domestic supply (i.e., designated MUN), unless certain exception criteria apply. Pursuant to these criteria, the Regional Board exempted certain waters, including Rattlesnake, Siphon, and Sand Canyon Reservoirs, San Diego Creek, and Newport Bay, from the MUN designation. **The Aliso Creek Hydrologic Sub-area is not exempted from the MUN designation under the San Diego Regional Water Board’s Basin Plan.**

Table 5 Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
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Table 5 Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
001-008, Storm-007 & 008	Irvine Groundwater Management Zone	<u>Present or Potential:</u> Municipal and domestic supply, agricultural supply, industrial service supply, and industrial process supply.
003	Rattlesnake Reservoir	<u>Present or Potential:</u> Agricultural supply, water contact recreation, non-contact water recreation, warm freshwater habitat, wildlife habitat Excepted from Municipal and Domestic Supply
004	Sand Canyon Reservoir	<u>Present or Potential:</u> Agricultural supply, water contact recreation, non-contact water recreation, warm freshwater habitat, wildlife habitat Excepted from Municipal and Domestic Supply
007	Siphon Reservoir	<u>Present or Potential:</u> Agricultural supply, water contact recreation, non-contact water recreation, warm freshwater habitat, wildlife habitat Excepted from Municipal and Domestic Supply
001 & 008	Aliso Creek Hydrologic Sub-area (Per San Diego Regional Board's Basin Plan)	<u>Present or Potential:</u> Municipal and domestic supply and agricultural supply.
Storm-007	San Diego Creek, Reach 1	<u>Present or Potential:</u> Water contact recreation, non-contact water recreation, warm freshwater habitat, and wildlife habitat Excepted from Municipal and Domestic Supply
Storm-008	San Joaquin Freshwater Marsh	<u>Present or Potential:</u> Water contact recreation; non-contact water recreation; warm freshwater habitat; preservation of biological habitats of special significance; rare, threatened or endangered species; and wildlife habitat Excepted from Municipal and Domestic Supply
Storm-007 & Storm-008	Newport Bay, Upper	<u>Present or Potential:</u> Navigation, Water contact recreation, Non-contact water recreation, Commercial and sport fishing, Preservation of biological habitats of special significance, Wildlife habitat, Rare, threatened or endangered species, Spawning, reproduction, and development, Marine habitat, Shellfish harvesting, and Estuarine habitat. Excepted from Municipal and Domestic Supply

Requirements of this Order implement the Basin **Plans of this Regional Water Board and the San Diego Regional Water Board.**

8. Order No. R8-2015-0024, Findings, page 13, replace Section II.V., as follows:

V. TMDLs. On October 11, 2011, the U. S. EPA issued its final decision regarding the State's 2010 Integrated Clean Water Act Sections 303(d) and 305(b) list of impaired water bodies. This list includes Reaches 1 and 2 of San Diego Creek and Upper and Lower Newport Bay. One or more of these water

bodies were determined to be impaired by one or more pollutants, including: fecal coliform, nutrients, metals and pesticides. Nutrient, fecal coliform, sediment, and toxic pollutants TMDLs have been established for the San Diego Creek/Newport Bay watershed. There is no discharge of treated wastewater or storm water to these impaired waterbodies under normal conditions (overflow from Sand Canyon is allowed only when the flow in San Diego Creek at Campus Drive is above 50 cubic feet per second that is under extreme storm conditions). **In addition, under emergency conditions and on a temporary basis the Discharger can discharge tertiary treated effluent to Reach 1 of San Diego Creek, which is tributary to Newport Bay, provided that several conditions are met and the total nitrogen and total phosphorous loads that are discharged and not excluded by the nutrient TMDL are offset at a ratio of 1.5:1.** This Order does implement relevant TMDL requirements (see Attachment F).

9. Order No. R8-2015-0024, Discharge Prohibitions, page 14, replace Section III.F.4., as follows:

4. For discharges of recycled water **from Sand Canyon Reservoir** to San Diego Creek or its tributaries ~~from the Facility~~ when the flow in San Diego Creek at Campus Drive is less than 50 cubic feet per second, the Discharger shall have an approved offset program to offset **any total nitrogen loads, discharged in excess of 1 mg/L total nitrogen concentration, and any total phosphorous loads** ~~nutrient discharged. discharges in excess of 1 mg/L total inorganic nitrogen.~~

10. Order No. R8-2015-0024, Effluent Limitations and Discharge Specifications, page 14 and 15, replace title and paragraph of Section IV.A., title of Section IV.A.1. and title of Table 6., as follows:

A. Tertiary Treated Effluent Limitations – Discharge Points 003, 004, & 007 & Storm-007

Unless otherwise specifically specified hereinafter, compliance with the following effluent limitations is measured at monitoring locations M-003A, M-004A, ~~and M-007~~, **and M-008** as described in the attached MRP (Attachment E).

1. Final Effluent Limitations – Discharge Point 003, 004, & 007 & Storm-007

- a. The Discharge shall maintain compliance with the following effluent limitations:

Table 6. Effluent Limitations at DPs 003, 004, & 007, Storm-007

11. Order No. R8-2015-0024, Effluent Limitations and Discharge Specifications, page 14, replace footnote No. 4, as follows:

Compliance with this limitation is to be achieved at M-003B, M-004B, ~~and M-007~~ and M-008 as described in the attached MRP (Attachment E.). Also, for M-008 see Section VII.O – Compliance Determination.

12. Order No. R8-2015-0024, Effluent Limitations and Discharge Specifications, page 15, replace Section IV.A.1.c., as follows:

- c. **TDS Limitations** – For DPs 003, 004 and 007 the 12-month flow weighted running average TDS constituent concentrations shall not exceed 720 mg/L and for Storm-007 the 12-month flow weighted running average TDS constituent concentrations shall not exceed 1,500 mg/L.

13. Order No. R8-2015-0024, Effluent Limitations and Discharge Specifications, page 16, add Section IV.A.1.f. and g., as follows:

- f. The Discharger shall offset the mass loads (in lbs/rainfall year) of Total Nitrogen (TN) and Total Phosphorous (TP) discharged, through Storm-007, to Reach 1 of San Diego Creek at a ratio 1.5:1 in accordance with the following conditions:

(1) When San Diego Creek flow is less than 50 cfs at Campus Drive TN mass loads discharged must be offset.

(2) TP mass loads discharged must be offset under any flow condition in San Diego Creek at Campus Drive.

- g. The Discharger shall quantify and report the TN and TP offset credits, in lbs/year, for every rainfall year (October-September) starting with the rainfall year of October 2017 through September 2018. The Discharger shall report the TN and TP removal credits available for offset in the annual monitoring report. The Discharger shall include in the offset report details regarding the location of TN and TP removal sites, inlet/outlet concentrations, pertinent dates, analytical methods used, equations used, map depicting the TN and TP removal locations, and load removal calculations in a tabulated format. TN and TP mass loads discharged to San Diego Creek must be offset with TN and TP mass removal credits accumulated within a three-rainfall year period that includes the rainfall year in which the discharge occurred.

14. Order No. R8-2015-0024, Effluent Limitations and Discharge Specifications, Reclamation Specifications, page 17, replace Table 7 and Section IV.C.1.b., as follows:

Table 7 Recycled Water Effluent Limitations at DPs 001, 002, & 008

Parameter	Units	Effluent Limitations			
		Average Monthly	Average Weekly	Instantaneous Minimum 12-month running average	This entire column to be deleted Instantaneous Maximum
Biochemical Oxygen Demand 5-day @ 20°C	mg/L	20	30	--	--
Total Suspended Solids	mg/L	20	30	--	--
pH	standard units	6 (instantaneous minimum)	9 (instantaneous maximum)	6- -	9
Chloride	mg/L			375	
Sulfate	mg/L			375	
% Sodium	%			60	
Nitrate	mg/L			45	
Iron	mg/L			0.3	
Manganese	mg/L			0.05	
MBAS	mg/L			0.5	
Boron	mg/L			0.	
Aluminum	mg/L			1.0	
Arsenic	mg/L			0.05	
Antimony	mg/L			0.006	
Barium	mg/L			1.0	
Beryllium	mg/L			0.004	
Cadmium	mg/L			0.005	
Cyanide	mg/L			0.2	
Mercury	mg/L			0.002	
Nickel	mg/L			0.1	
Selenium	mg/L			0.05	
Thallium	mg/L			0.002	

Note: The 12-month running average effluent limitations on this table apply only to recycled water use that overlie the Aliso Creek Hydrologic Sub-area within the San Diego Regional Water Board jurisdictional boundary.

- b. TDS Limitations: For recycled water use on sites overlying the Irvine Groundwater Management Zone, the 12-month flow weighted running average TDS concentration shall not exceed 910 mg/L. **For recycled water use on sites overlying the Aliso Creek Hydrologic Sub-area, the 12-month running average TDS concentrations shall not exceed 1,000 mg/L.**

15. Order No. R8-2015-0024, Effluent Limitations and Discharge Specifications, Reclamation Specifications, page 21, add Section IV.C.10. and Section IV.C.11., as follows:

10. Uses of recycled water with frequent or routine application (for example: agricultural or landscape irrigation uses) shall be at agronomic rates and shall consider soil, climate, and plant demand. In addition, application of recycled water and use of fertilizers shall be at a rate that takes into consideration nutrient levels in recycled water and nutrient demand by plants. This Regional Water Board may require the Discharger to submit an Implementation or Operations and Management Plan specifying agronomic rates and nutrient application for the use area(s) and a set of reasonably practicable measures to ensure compliance with this Order. The Discharger must implement the following to ensure that recycled water and fertilizer are applied in use sites at agronomic rates:

- a. Monitor nutrient concentrations in recycled water supplies and notify recycled water site supervisors of the nutrient concentrations of recycled water.**
- b. Conduct periodic inspections of end use sites to verify the agronomic application rates of recycled water considering fertilizer application and the nutrient level of the recycled water applied to the landscape or provide monthly water budgets to each use site based on agronomic rates, provide monthly reporting to each use site of actual use compared with the agronomic water budget for each use site, and target sites with recycled water consistently above the agronomic rate for follow-up.**

11. In lieu of meeting the recycled water effluent limit for total nitrogen of 10 mg/L (Section IV.C.1.c., Table 7), the Discharger may elect to conduct a nitrogen study to verify that the use of recycled water from IRWD's Facilities for landscape or agricultural irrigation does not cause the water quality of the underlying groundwater to exceed the groundwater quality objective for nitrate of 45 mg/L (as nitrate) nor exacerbate an existing condition that exceeds the nitrate groundwater quality objective. The nitrogen study must evaluate factors such as existing nitrogen removal achieved at the plant, need for additional treatment processes to remove nitrogen, fate and transport of nitrogen in the groundwater, groundwater monitoring, application of recycled water and fertilizer at agronomic rates, nitrogen uptake by turf grass or other vegetation, and other best management practices. A workplan for the nitrogen study, if needed, must be received at our Santa Ana Regional Water Board's office by June 15, 2019, and must identify proposed tasks and milestones for completing the nitrogen study, and a schedule for

completing the study activities. A nitrogen study report documenting the results of the study must be received at this Regional Water Board by March 15, 2020. For a valid reason, the Executive Officer may grant additional time to submit the workplan and/or report for the nitrogen study.

16. Order No. R8-2015-0024, Provisions, page 28, add Section VI.C.2.e., as follows:

e. By December 31, 2018, the Discharger shall submit for approval by the Executive Officer, a TN and TP Offset plan that details the manner in which the Discharger will gather all pertinent information to document the removal credits of TN and TP loads in the NTS and San Joaquin Freshwater Marsh managed by IRWD and the reporting of the data as required in this Order. For a valid reason, the Executive Officer may grant additional time to submit the TN and TP Offset plan.

17. Order No. R8-2015-0024, Provisions, Other Special Provisions, page 33, add Section VI.C.6.b through d., as follows:

b. Before tertiary treated effluent is discharged to Reach 1 of San Diego Creek, through Discharge Point (DP) Storm-007, the Discharger shall give this Regional Water Board and the California Department of Fish and Wildlife (Carlsbad Office) at least 48-hour advance notice regarding their intent to discharge and fulfill the following conditions:

- (1) Exhaust its recycled water storage capacity;**
- (2) Stop importing water purchased from the Metropolitan Water District to augment its recycled water system;**
- (3) Shutoff all irrigation water wells providing water to its recycled water system;**
- (4) Stop producing recycled water at Los Alisos Water Recycling Plant;**
- (5) Send recycled water to the Orange County Water District (OCWD) through the Green Acres Project;**
- (6) Dispose of recycled water, to the maximum extent possible, through the Orange County Sanitation District's (OCSA's) ocean outfall;**
- (7) If possible, shutoff all Department of Justice wells that deliver water into the recycled water system, while still complying with the Record of Decision with the Department of the Navy pertaining to the operation of these wells;**
- (8) Send all allowable recycled water (subject to physical conveyance capacities) to South Orange County Wastewater Authority's ocean outfall;**

(9) Confirm and document that OCWD and OCSD cannot take additional recycled water flow; and

(10) If a diversion of sewage is feasible from Michelson WRP to OCSD, confirm that OCSD cannot receive raw sewage flow from IRWD MWRP's service area.

c. If a diversion of sewage is feasible from Michelson WRP to OCSD, IRWD shall maintain real-time communication and coordination with OCSD staff so that IRWD can resume, as-soon-as-possible, the disposal of raw sewage from IRWD MWRP's service area through OCSD treatment plants and discontinue the discharge of tertiary treated effluent to Reach 1 of San Diego Creek through DP Storm-007.

d. The Discharger must develop and implement a rainy season discharge contingency plan that includes details describing the measures that IRWD will take to operate the Michelson WRP in an enhanced nutrient removal mode to produce recycled water with an average total nitrogen and total phosphorous concentrations of 10 mg/L and 0.3 mg/L respectively, as operational goals, and comply with the total chlorine residual effluent limits (amongst other parameters) when discharging tertiary treated effluent through DP Storm-007. This plan must include pertinent details regarding the operational strategies, additional equipment and resources, chemicals, timing of implementation, testing, power loss backup equipment, planning and preparation, and OCSD/OCWD/IRWD staff coordination and contact list (emails, phone numbers, and functions). The rainy season discharge contingency plan must be made available to Regional Water Board staff upon request.

18. Order No. R8-2015-0024, Compliance Determination, page 35, replace Section VII.I., as follows:

I. Total Chlorine Residual Limitation (TCR)

For Rattlesnake and Sand Canyon Reservoirs the compliance determination for total chlorine residual shall be determined at the boundary of the zone of initial dilution approved by the Regional Board Executive Officer, ~~except for~~ **and for** Syphon Reservoir, ~~which~~ is determined at the discharge point. **For tertiary treated effluent discharged through Discharge Point Storm-007 the compliance determinations for total chlorine residual shall be based on 99% compliance. To determine 99% compliance with the effluent limitation for total chlorine residual, the following conditions shall be satisfied:**

1. For TCR Limit specified in Section IV.A.1.:

- a. The total time during which the total chlorine residual values are above 0.1 mg/L (instantaneous maximum value) shall not exceed 7 hours and 26 minutes in any calendar month;
- b. No individual excursion from 0.1 mg/L value shall exceed 5 minutes; and
- c. No individual excursion shall exceed 5.0 mg/L.

19. Attachment C, Flow Schematic, page C-1, replace page C-1 of Attachment C with page C-1 that is attached to this Order.

20. Attachment E, Monitoring Locations, page E-7, replace Table 2, as follows:

Table 2 Monitoring Station Locations

Discharge Point Name	Monitoring Location Name	Monitoring Location Description	Latitude and Longitude
--	M-INF 1	Influent distribution channel at MWRP	33°39'48" & 117°50'19"
	M-INF 2	Influent channel at LAWRP	33°38'16" & 117°42'54"
001	REC-001	Effluent from MWRP to recycling system	33°39'50" & 117°50'17"
002	REC-002	Effluent to San Joaquin Reservoir	33°37'12" & 117°50'39"
003	M-003A	Effluent to Rattlesnake Reservoir	33°43'37" & 117°44'43"
003	M-003B	Rattlesnake Reservoir ZID boundary	33°43'37" & 117°44'27"
004	M-004A	Effluent to Sand Canyon Reservoir	33°38'55" & 117°47'47"
004	M-004B	Sand Canyon Reservoir ZID boundary	33°38'50" & 117°47'50"
007	M-007	Effluent to Syphon Reservoir	33°42'36" & 117°43'56"
008	REC-003	Effluent from LAWRP to recycling system	33°38'16" & 117°42'50"
Storm-007	M-008	Emergency stormwater overflow or tertiary treated effluent discharge to reach Reach 1 of San Diego Creek	33°39'47" & 117°50'16"
Storm-008	M-009	Emergency stormwater overflow to San Joaquin Marsh	33°39'51" & 117°50'27"
	R-001	Receiving surface water—Rattlesnake Reservoir 3 location spatial composite	33°43'46" & 117°44'12" 33°43'42" & 117°44'23" 33°43'41" & 117°44'29"
	R-002	Receiving surface water—Sand Canyon Reservoir location 3 spatial composite	33°38'38" & 117°47'46" 33°38'48" & 117°47'48" 33°38'51" & 117°47'47"
	R-003	Receiving surface water—Syphon Reservoir location 3 spatial composite	33°43'39" & 117°44'28" 33°43'43" & 117°44'20" 33°43'46" & 117°44'13"
	R-004U	Upstream monitoring location in San Diego Creek, 100 meters upstream from Discharge Point Storm-007	33°39'48" & 117°50'15"
	R-004D	Downstream monitoring location in San Diego Creek, within 500 meters downstream from Discharge Point Storm-007	33°39'43" & 117°50'20"

21. Attachment E, Effluent Monitoring Requirements to Surface Waters, page E-9 and E-10, replace the title of Section IV.A., paragraph IV.A.1., and title of Table 4, as follows:

A. Effluent Monitoring Locations M-003, M-004, & M-007 & M-008

1. The Discharger shall monitor tertiary treated effluent for DP-003, DP-004, DP-007 and Storm-007 at Monitoring Locations M-003A, M-004A, and M-007 **and M-008, respectively**, as follows.

Table 4 Tertiary Effluent Monitoring M-003A, M-004A, & M-007 & M-008

22. Attachment E, Effluent Monitoring Requirements to Surface Waters, page E-12, replace Section IV.A.2. and Table 5, as follows:

2. The Discharger shall monitor tertiary treated effluent for DP 003, DP 004, ~~and DP 007~~ **and Storm-007** at Monitoring Locations M-003B, M-004B, ~~and M-007~~ **and M-008**, respectively, for Total Residual Chlorine **and Total Dissolved Solids** as follows.

Table 5 Tertiary Effluent Monitoring M-003B, M-004B, & M-007 & M-008*

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method and Minimum Level, units, respectively
Total Residual Chlorine	mg/L	Grab	Weekly	See Section I.A.3. above of this MRP
Total Residual Chlorine*	mg/L	Recorder	Continuous	--
Total Dissolved Solids*	mg/L	Composite	Monthly	See Section I.A.3. above of this MRP

Note: * = Apply only at M-008 when discharging tertiary treated effluent.

23. Attachment E, Effluent Monitoring Requirements to Surface Waters, page E-13, replace Sections IV.A.5. and IV.A.7. as follows:

5. If there are no discharges of **tertiary** treated effluent to DP 003, DP 004, DP ~~and 007~~ **and Storm-007**, the Discharger is not required to monitor for all the constituents in Table 4 and Table 5, above. However, the Discharger shall record on a permanent log on a daily basis when there is no discharge and report this information together with the monthly report.
7. The Discharger may demonstrate compliance with the Effluent Limitations (except for Total Residual Chlorine **and Total Dissolved Solids at M-008**), Discharge Specifications, and Monitoring and Reporting Requirements in this Order for **tertiary treated effluent** discharges at Discharge Locations DP 002, DP 003, DP 004, ~~and DP 007~~ **and Storm-007** by monitoring at Discharge Location DP 001 (monitoring location REC-001).

24. Attachment E, Whole Effluent Toxicity Testing Requirements, page E-13, replace the title of Section V.A., as follows:

A. Toxicity Monitoring Requirements at M-003A, M-004A, and M-007 and M-008

25. Attachment E, Recycled Water Monitoring Requirements, page E-16, replace Table 6, as follows:

Table 6 Reclamation Monitoring at REC-001, REC-002, & REC-003

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Flow	mgd	Recorder/Totalizer	Continuous	---
pH	Standard units	Recorder/Totalizer	Continuous	--
CT ¹⁴	mg/L-min	Recorder	Continuous	See Section I.A.3., above, of this MRP
Turbidity	NTU	Recorder	Continuous	"
Coliform Organisms	MPN per 100 mL	Grab	Daily	"
BOD ₅	mg/L	Composite	Weekly	See Section I.A.3., above, of this MRP "
Total Suspended Solids	mg/L	Composite	Weekly	See Section I.A.3., above, of this MRP "
TDS	mg/L	Composite	Monthly	See Section I.A.3., above, of this MRP "
Chloride	mg/L	Composite	Quarterly	"
Percent Sodium	%	Composite	Quarterly	"
Sulfate	mg/L	Composite	Quarterly	"
Iron	mg/L	Composite	Quarterly	"
Manganese	mg/L	Composite	Quarterly	"
MBAS	mg/L	Composite	Quarterly	"
Boron	mg/L	Composite	Quarterly	"
Aluminum	mg/L	Composite	Annually	"
Barium	mg/L	Composite	Annually	"
Nitrate-Nitrogen	mg/L	Composite	Quarterly	"
Total Nitrogen	mg/L	Composite	Quarterly	"
Priority Pollutants	mg/L	Composite	Annually	"

26. Attachment E, Recycled Water Monitoring Requirements, page E-16, add Section VII.A.2., as follows:

2. The Discharger shall increase the sampling frequency from weekly to daily, from quarterly to monthly, and from annually to quarterly for any constituent that exceeds the recycled water effluent limitations specified in Table 7 of this Order. To return to the monitoring frequency specified, the Discharger shall request and receive approval from the Regional

Water Board's Executive Officer or designee after maintaining compliance with the effluent limitations in question for at least three consecutive monitoring periods.

27. Attachment E, Recycled Water Monitoring Requirements, page E-16, replace Section VII.B., as follows:

B. Monitoring Users

~~Whenever recycled water is supplied to a user, the Discharger shall record on a permanent log: the volume of recycled water supplied; the user of recycled water; the locations of those sites including the names of the groundwater management zones underlying the recycled water use sites; type of use (e.g. irrigation, industrial, etc); and the dates at which water is supplied. The Discharger shall submit annually a summary report of the recorded information by groundwater management zone to the Regional Water Board.~~ **The Discharger shall record the information described below and shall submit annually a summary report of the recorded information by groundwater management zone to the Regional Water Board.**

- 1. Use site information summary that includes the following: a) Name and address of use site; b) Owner of the site; c) Name and phone number of the onsite supervisor; d) Groundwater management zones underlying the use sites; e) Annual volume of recycled water delivered; f) Total area (in acres) of each landscape irrigation site; g) The amount of nitrogen (in pounds per acre per year) applied in recycled water on each landscape irrigation site (required if the Discharger elects to implement a nitrogen study per Section IV.C.11. of the Order); and h) Type of use (e.g. irrigation, industrial, etc.).**
- 2. Recycled water use site inspection summary that includes the number of inspections conducted by the Discharger and number of violations, if any.**
- 3. Summary of compliance status and enforcement actions taken by the Discharger for recycled water users known to be in violation of its rules and regulations, which include a description of violations (cause, duration, and impact), measures taken to correct and eliminate recurrence of a violation or threatened violation, and compliance schedules issued to use sites.**

28. Attachment E, Receiving Water Monitoring Requirements – Surface Water and Groundwater, page E-17, replace the paragraph of Section VIII.A.1. and title of Table 7, as follows:

- 1. The Discharger shall monitor receiving water at R-001, R-002, & R-003, R-004U & R-004D as follows:**

Table 7 Receiving Water Monitoring Requirements at R-001, R-002, & R-003, R-004U & R-004D

29. Attachment F, List of Tables, page F-2, replace the List of Tables, as follows:

List of Tables

Table 1.	Facility Information	21
Table 2.	Facilities' Treatment Processes	Error!
Bookmark not defined.		
Table 3.	Recycled Water Produced	Error!
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Table 4.	Closed Tanks	Error!
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Table 5.	Open Reservoirs	Error!
Bookmark not defined.		
Table 6.	Treated Water Sent to OCWD and OCSD	Error!
Bookmark not defined.		
Table 7.	Summary of Discharge Points and Receiving Waters	25
Table 8.	Historic Effluent Monitoring Data at DPs 003, 004 and 007	Error!
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Table 9.	Compliance Summary	Error!
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Table 10.	Basin Plan Beneficial Uses	29
Table 11.	Applicable Basin Plan Surface Water Quality Objectives	Error!
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Table 12.	TDS/ Ammonia-Nitrogen Limits	Error!
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Table 13.	MWRP Effluent Monitoring Data, µg/L	Error!
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Table 14.	RPA Evaluation for DPs 003, 004, & 007	Error!
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Table 15.	Tertiary Effluent BOD ₅ and TSS Limits	Error!
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Table 16.	Summary of Water Quality-based Final Effluent Limitations for Discharge Points 003, 004, & 007	34
Table 17	Recycled Water Effluent Limitations at DPs 001, 002, & 008	F-34
Table 18 17.	Stormwater Discharge Limits at Storm-007 & Storm-008	Error!
Bookmark not defined.		

30. Attachment F, Permit Information, page F-4, replace Table 1, as follows:

Table 1. Facility Information

WDID	8 302006002
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Discharger/Operator	Irvine Ranch Water District	
Names of Facilities	Michelson Water Recycling Plant	Los Alisos Water Recycling Plant
Addresses	3512 Michelson Drive	22312 Muirlands Boulevard
	Irvine, CA 92612	Lake Forest, CA 92630
	Orange County	
Facility Contact, Title and Phone	Gaspar Garza Randy C. Lee, Assistant Director of Recycling Operations Manager , Phone: (949) 453-5780 5826	
Authorized Person to Sign and Submit Reports	Jose Zepeda Patrick Shields, Executive Director of Recycling Operations, Phone: (949) 453-5720 5572	
Mailing Address	15600 Sand Canyon Avenue, Irvine, CA 92618	
Billing Address	P.O. BOX 57000, Irvine, CA 92619-7000	
Type of Facility	POTW	
Major or Minor Facility	Major	
Threat to Water Quality	1	
Complexity	A	
Pretreatment Program	Y	
Reclamation Requirements	Producer/User	
Facility Permitted Flow	28 million gallons per day (mgd)	7.5 mgd - secondary treatment and 5.5 mgd – tertiary treatment
Type of Permitted Flows	NPDES	WDR (Recycled Water only)
Facility Design Flow	28 mgd	7.5 mgd - secondary treatment and 5.5 mgd – tertiary treatment
Watershed	San Diego Creek/Newport Bay watershed	San Diego Creek/Newport Bay watershed
Receiving Water	Irvine Groundwater Management Zone. Aliso Creek Hydrologic Sub-area (Per San Diego Regional Board's Basin Plan) Surface waters: Rattlesnake, Syphon, and Sand Canyon Reservoirs, Reach 1 of San Diego Creek, San Joaquin Freshwater Marsh and Upper Newport Bay.	
Receiving Water Type	Inland Surface Water, Estuary and Groundwater	

31. Attachment F, Permit Information, page F-5, replace Section I.B., as follows:

- B.** ~~The MWRP discharges wastewater to waters of the United States, including: 1) tertiary treated recycled water to Sand Canyon Reservoir, Syphon Reservoir, and Rattlesnake Reservoir; 2) emergency pumped storm water to Reach 1 of San Diego Creek and San Joaquin Freshwater Marsh, which are tributaries of Upper Newport Bay.~~ **The MWRP discharges wastewater to waters of the United States, including: 1) tertiary treated effluent, for recycled water storage purposes, to Sand Canyon Reservoir, Syphon Reservoir, and Rattlesnake Reservoir; 2) under emergency conditions, pumped storm water or tertiary treated effluent (as a last resort disposal alternative) to Reach 1 of San Diego Creek; and 3) under emergency conditions, pumped storm water to San Joaquin Freshwater Marsh. Both Reach 1 of San Diego Creek and the San Joaquin Freshwater Marsh** ~~which are tributaries of Upper Newport Bay.~~ The discharge from MWRP is

currently regulated by Order No. R8-2007-0003, which was adopted on November 30, 2007 and expired on November 1, 2012. Order No. R8-2007-0003 was amended by Order No. R8-2008-0072 on September 5, 2008. The amendment corrected the discharge prohibition period of Sand Canyon Reservoir, modified reclamation specifications, and provided flexibility to demonstrate compliance at monitoring location REC-001 for other discharge points. Also, on June 15, 2012 Order No. R8-2007-0003 was amended by Order No. R8-2012-0004, which added Syphon Reservoir as an additional discharge point for recycled water storage in the distribution system. The terms and conditions of the current Order and its amendment have been automatically continued and remain in effect until new Waste Discharge and Water Reclamation Requirements and NPDES permit are adopted pursuant to this Order.

The LAWRP discharges tertiary treated and disinfected recycled water to its recycled water distribution system within the Lake Forest service area and/or to IRWD's Zone B. **The discharge of recycled water, produced at the Facilities, within the City of Lake Forest are made to use areas located within the jurisdiction of this Regional Water Board and the San Diego Regional Water Board, however, as explained in Section I.C below, this Regional Water Board has been designated to be the sole Regional Water Board that regulates the treatment and distribution of recycled water produced at the IRWD's Facilities.** When discharging to IRWD's Zone B the recycled water is not discharged to IRWD's open reservoirs that are waters of the US. On the other hand, recycled water produced at MWRP may be received at LAWRP recycled water pump station's wetwell, on a seasonal basis, for distribution within the Lake Forest service area of LAWRP. The recycled water discharge from this facility is currently regulated under Order No. 94-3 issued to the South Orange County Reclamation Authority (SOCRA) by this Regional Water Board on March 11, 1994. SOCRA changed its name in 2001 to the South Orange County Wastewater Authority (SOCWA). Also, Order No. 94-3 regulates recycled water discharges from El Toro Water District's Water Recycling Plant. Both IRWD and El Toro Water District are members of SOCWA. When recycled water from MWRP is received at LAWRP or when a portion of its secondary treated effluent is not reclaimed, LAWRP's undisinfected secondary effluent is discharged to the Pacific Ocean through the Aliso Creek ocean outfall operated by SOCWA. The discharge to the Pacific Ocean is regulated by the San Diego Regional Water Quality Control Board through Order No. R9-2012-0013 issued to SOCWA.

32. Attachment F, Permit Information, page F-6, add a fourth and fifth paragraph at the end of Section I.C., as follows:

The San Diego Regional Water Quality Control Board (San Diego Regional Water Board) regulates the use of recycled water, produced at the Discharger's Facilities, within the Aliso Creek Hydrologic Sub-area, of its region, under Order 97-52, Waste Discharge and Water Recycling

Requirements for the Production and Purveyance of Recycled Water by Member Agencies of the South Orange County Reclamation Authority, which is a regional water recycling permit. In a letter dated March 30, 2016, the Discharger requested to this Regional Water Board and the San Diego Regional Water Board the designation of this Regional Water Board as the sole Regional Water Board for the regulation of the treatment, distribution and use of recycled water produced at the Discharger's water recycling plants. Including the distribution of recycled water to use areas located within the boundaries of the San Diego Regional Water Board in the City of Lake Forest. The Discharger made the regional water board designation request pursuant to Section 13228(a) of the California Water Code. The San Diego and this Regional Water Boards agreed with IRWD's request to designate this Regional Water Board as the sole Regional Water Board for regulating the IRWD's Facilities. The San Diego and this Regional Water Boards memorialized the designation agreement in separate letters addressed to the Discharger, which were dated July 22, 2016 and September 19, 2016, respectively, and this designation will become effective once the San Diego Regional Water Board amends Order 97-52 or adopts a new master reclamation permit for the South Orange County Wastewater Authority that excludes IRWD's.

On July 18, 2017 the Discharger submitted a Report of Waste Discharge requesting authorization to temporarily discharge tertiary treated and disinfected effluent, under emergency conditions, to Reach 1 of San Diego Creek through Discharge Point Storm-007 (Storm-007). The Discharger submitted several documents in support of the proposed discharge and in response to information request from Regional Water Board staff to demonstrate that the proposed discharge enhances and does not impact the water quality of Reach 1 of San Diego Creek and Newport Bay and complies with the Water Quality Control Policy for the Enclosed Bays and Estuaries of California, as amended by SWRCB's Resolution No. 95-84, and the conditions of the Nutrient Total Maximum Daily Load (TMDL) and other TMDLs (fecal coliform, sediment and toxic pollutants) established for the San Diego Creek/Newport Bay watershed. Regional Water Board staff requested and received additional information from the Discharger until June 6, 2018.

33. Attachment F, Facility Description, page F-12, replace Section II.B.1, as follows:

1. Discharge Points for Tertiary Treated Effluent and ~~Recycling~~ Recycled Water Reuse

Recycled water is delivered via DP-001 and DP-008 to the distribution system serving recycled water users.

Recycled water produced at MWRP is delivered via DP-002 to San Joaquin Reservoir for storage.

For recycled water storage purposes, Tertiary tertiary treated effluent from recycled water produced at MWRP is dechlorinated and discharged to surface waters at three points located at Rattlesnake Reservoir, Sand Canyon Reservoir, and Syphon Reservoir, designated as DP-003, DP-004, and DP-007, respectively. Tertiary treated recycled water produced at LAWRP and discharged through DP-008 is not discharged to the reservoirs that are filled with recycled water produced at MWRP. Also, under emergency conditions and as last resort for disposal, tertiary treated effluent may be temporarily discharged to Reach 1 of San Diego Creek through Storm-007, which is tributary to Upper Newport Bay.

34. Attachment F, Facility Description, page F-12, replace Table 7, as follows:

Table 7. Summary of Discharge Points and Receiving Waters

Discharge Serial No.	Latitude	Longitude	Description and Receiving Waters	Flow & Frequency
001	33°39'50"N	117°50'17"W	Recycled water from MWRP to distribution system serving recycled water users, overlying Irvine GMZ	12.3 mgd Continuous
002	33°37'12"N	117°50'39"W	Recycled water from MWRP to San Joaquin Reservoir, overlying Irvine GMZ	Variable up to 12.3 mgd
003	33°43'37"N	117°44'27"W	Tertiary treated effluent from MWRP to Rattlesnake Reservoir, overlying Irvine GMZ	Variable up to 12.3 mgd
004	33°38'55"N	117°47'47"W	Tertiary treated effluent from MWRP to Sand Canyon Reservoir, overlying Irvine GMZ	Variable up to 12.3 mgd
007	33°42'36"N	117°43'56"W	Tertiary treated effluent from MWRP to Syphon Reservoir, overlying Irvine GMZ	Variable up to 12.3 mgd
008	33°38'16"N	117°42'50"W	Recycled water from LAWRP and/or MWRP to distribution system serving recycled water users, overlying Irvine GMZ & Aliso Creek Hydrologic Sub-area (Per San Diego Regional Board's Basin Plan)	Variable up to 5.5 mgd
Storm-007	33°39'47"N	117°42'50"W	Pumped stormwater (emergency) or tertiary treated effluent (emergency) from MWRP to San Diego Creek Reach 1 tributary to Upper Newport Bay	Unmetered
Storm-008	33°39'51"N	117°50'27"W	Pumped stormwater (emergency) from MWRP to San Joaquin Freshwater Marsh tributary to Upper Newport Bay	Unmetered

35. Attachment F, Summary of Existing Requirements and Self-Monitoring Report (SMR) Data, page F-14, replace the last paragraph of Section II.C.1., as follows:

The previous Order prohibited the direct discharge of tertiary treated recycled water to surface water, except into the three reservoirs, Rattlesnake Reservoir, Syphon Reservoir, and Sand Canyon Reservoir. However, during wet weather

conditions, storm water runoff from the reservoir watershed occasionally causes overflows¹ from Sand Canyon Reservoir into San Diego Creek. The previous Order R8-2007-0003 allowed emergency discharge of water from the reservoir during heavy rainfall events. The Discharger has implemented best management practices to minimize overflows from Sand Canyon Reservoir by limiting recycled water storage during the rainy season. During the term of Order No. R8-2007-0003, Sand Canyon Reservoir has overflowed twice in 2010 and 2011 during the rainy season. This Order permits overflows from Sand Canyon Reservoir under certain storm conditions when the flow in San Diego Creek at Campus Drive is expected to be 50 cubic feet per second or higher. Under such flow conditions, the discharge should have no impact on nutrient loading to San Diego Creek or Newport Bay. **Also, under emergency conditions and as last resort for disposal, tertiary treated effluent may be temporarily discharged to Reach 1 of San Diego Creek through Storm-007, which is tributary to Upper Newport Bay.** The Discharger must implement a total nitrogen (TN) offset program if the discharge **of recycled water, stormwater or tertiary treated effluent** occurs when the flow in San Diego Creek at Campus Drive is less than 50 cubic feet per second **and the TN concentration in the effluent exceeds 1 mg/L. Total phosphorous (TP) loads discharged must be offset regardless of the flow regime in San Diego Creek. IRWD operates a network of natural treatment wetlands in the San Diego Creek Watershed that remove nitrogen and phosphorus from San Diego Creek and its tributaries.**

36. Attachment F, Planned Changes, page F-16, replace the last paragraph of Section II.E., as follows:

IRWD is constructing a Biosolids Handling and Energy Recovery Facility at the MWRP site, which would treat wastes from both Facilities, and includes the construction and/or installation of sludge thickening equipment, anaerobic digesters, centrifuges for sludge dewatering, sludge drying equipment, anaerobically digestible material receiving station, and microturbines for generation of electricity and heat. The microturbines would operate with biogas produced in the anaerobic digesters. IRWD anticipates completion of the Biosolids Handling and Energy Recovery Facility by the end of 2016 **mid-2020. In addition, IRWD is pursuing the expansion of its recycled water storage capacity at Syphon Reservoir to almost 10 times the current reservoir capacity. IRWD is also working towards developing an emergency discharge point to Reach 1 of the Santa Ana River for tertiary treated effluent.**

37. Attachment F, Applicable Plans, Policies, and Regulations, page F-16, replace Section III.A., as follows:

A. Legal Authorities

This Order is issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. Environmental

Protection Agency (USEPA) and Chapter 5.5, Division 7 of the California Water Code (commencing with Section 13370). It shall serve as a NPDES permit for point source discharges from this Facility to surface waters. This Order also serves as Waste Discharge Requirements (WDRs) pursuant to article 4, Chapter 4, Division 7 of the Water Code (commencing with Section 13260) and as a master reclamation permit pursuant to Section 13523.1 of Article 4, Chapter 7, Division 7 of the California Water Code. This Order, as a master reclamation permit, includes Producer/User Reclamation Requirements to regulate recycled water use for irrigation and other industrial uses. **In addition, this Order includes reclamation specification for the use of recycled water produced at IRWD's Facilities at use areas located within the jurisdictional boundaries of the San Diego Regional Water Board, in the City of Lake Forest, pursuant to a Regional Water Board designation request made by IRWD pursuant to Section 13228(a) of the California Water Code. The San Diego and this Regional Water Boards agreed with IRWD's request to designate this Regional Water Board as the sole Regional Water Board for regulating the IRWD's Water Recycling Plants and this designation will become effective once the San Diego Regional Water Board amends Order 97-52 or adopts a new master reclamation permit for the South Orange County Wastewater Authority that excludes IRWD.**

38. Attachment F, Applicable Plans, Policies, and Regulations, page F-17, replace Section III.B., as follows:

B. California Environmental Quality Act (CEQA)

Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of CEQA, Public Resources Code section 21000 et seq. (*County of Los Angeles v. California State Water Resources Control Board* (2006) 143 Cal.App.4th 985, mod. (Nov. 6, 2006, B184034) 50 Cal.Rptr.3d 619, 632-636.) For the plant expansion project, a mitigated negative declaration and addendum was adopted on July 2, 2003 and October 5, 2005, respectively. This action also involves the re-issuance of waste discharge requirements for an existing facility that discharges treated wastewater to land and as such, is exempt from the provisions of California Environmental Quality Act (commencing with Section 21100) in that the activity is exempt pursuant to Title 14 of the California Code of Regulations Section 15301". MWRP has completed the capacity expansion project Environmental Impact Report, SCH 2005051174 on December 28, 2005. The EIR was certified on February 27, 2006. The CEQA documents for discharge location DP-007, Syphon Reservoir, were completed on December 10, 2009 and a negative declaration was certified on December 14, 2009. The Final Supplemental Environmental Impact Report (SEIR) No. 1 for the MWRP Phase 2 and 3 Capacity Expansion Project, Biosolids Handling Component, SCH 2011031091, was certified on October 23, 2012. The SEIR No. 1 covered the CEQA process for the Biosolids Handling and Energy Recovery

Facility and includes the mitigation of potential odor sources with an adequate odor control system, monitoring for presence of endangered species and/or installation of noise control measures during nesting season, and considers measures to protect human health and safety during construction. **The CEQA document, Addendum No. 4 to the Michelson Water Reclamation Plant Phase 2 and 3 Capacity Expansion Project (Final EIR), addresses the temporary emergency discharge of tertiary treated effluent through Discharge Point Storm-007 to Reach 1 of San Diego Creek and was completed on March 2018 and a negative declaration was certified by IRWD on March 26, 2018.** Regional Board staff has reviewed the CEQA documents related these projects and finds that if the Discharger complies with the requirements specified in this Order, it should not have any significant impact on the water quality of the receiving water bodies.

39. Attachment F, Applicable Plans, Policies, and Regulations, page F-17, replace Section III.C.1. and Table 10, as follows:

1. **Water Quality Control Plans.** The Regional Water Board adopted an updated Water Quality Control Plan for the Santa Ana Basin (hereinafter Basin Plan) that became effective on January 24, 1995. The Basin Plan designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters addressed through the plan. On January 22, 2004, the Regional Water Board adopted Resolution No. R8-2004-0001, amending the Basin Plan to incorporate revised boundaries for groundwater subbasins, now termed “management zones”, new nitrate-nitrogen and TDS objectives for the new management zones, and new nitrogen and TDS management strategies applicable to both surface and ground waters. The State Water Board and Office of Administrative Law (OAL) approved the N/TDS Amendment on September 30, 2004 and December 23, 2004, respectively. EPA approved the surface water standards components of the N/TDS Amendment on June 20, 2007. Effluent limitations for TDS and TIN in this Order are based on applicable water quality objectives, adopted and now in effect as part of the N/TDS Amendment. TDS limits for recycled water use are based on the TDS objective for the affected groundwater management zone.

The San Diego Regional Water Board adopted the Water Quality Control Plan for their region on September 8, 1994 and subsequently approved by the State Water Resources Control Board (SWRCB) on December 13, 1994. Subsequent revisions to the San Diego Regional Water Board’s Basin Plan have also been adopted by their board and approved by the SWRCB. The San Diego Regional Water Board’s Basin Plan contains beneficial uses and water quality objectives. The reclamation specifications for recycled water use, within the Aliso Creek Hydrologic Sub-area, of this Order are consistent with the San Diego Regional Water Board’s Basin Plan.

In addition, State Water Resources Control Board (State Water Board) Resolution No. 88-63 (Sources of Drinking Water Policy) requires that, with certain exceptions, the Regional Water Board assign the municipal and domestic water supply use to water bodies. Based on the exception criteria specified in Resolution No. 88-63, the Regional Water Board excepted Reach 1 and 2 of the San Diego Creek, Rattlesnake, Syphon, and Sand Canyon Reservoirs, and Upper and Lower Newport Bay from the municipal and domestic supply beneficial use. **The Aliso Creek Hydrologic Sub-area is not exempted from the MUN designation under the San Diego Regional Water Board's Basin Plan.**

As previously discussed, the Facility discharges wastewater from several discharge points. The designated beneficial uses of receiving waters affected by the discharge from the Facility are as follows:

Table 10. Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
001 to 08	Irvine Groundwater Management Zone	<u>Present or Potential:</u> Municipal and domestic supply, agricultural supply, industrial service supply, and industrial process supply.
003	Rattlesnake Reservoir	<u>Present or Potential:</u> Agricultural supply, water contact recreation, non-contact water recreation, warm freshwater habitat, and wildlife habitat Excepted from Municipal and Domestic Supply
004	Sand Canyon Reservoir	<u>Present or Potential:</u> Agricultural supply, water contact recreation, non-contact water recreation, warm freshwater habitat, and wildlife habitat Excepted from Municipal and Domestic Supply
007	Syphon Reservoir	<u>Present or Potential:</u> Agricultural supply, water contact recreation, non-contact water recreation, warm freshwater habitat, and wildlife habitat Excepted from Municipal and Domestic Supply
001 & 008	Aliso Creek Hydrologic Sub-area (Per San Diego Regional Board's Basin Plan)	<u>Present or Potential:</u> Municipal and domestic supply and agricultural supply.
Storm-007	San Diego Creek, Reach 1	<u>Present or Potential:</u> Water contact recreation, non-contact water recreation, warm freshwater habitat, and wildlife habitat Excepted from Municipal and Domestic Supply
Storm-008	San Joaquin Freshwater Marsh	<u>Present or Potential:</u> Water contact recreation; non-contact water recreation; warm freshwater habitat; preservation of biological habitats of special significance; rare, threatened or endangered species; and wildlife habitat Excepted from Municipal and Domestic Supply

Table 10. Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
Storm-007 & Storm-008	Newport Bay, Upper	<u>Present or Potential:</u> Water contact recreation, Non-contact water recreation, Commercial and sport fishing, Preservation of biological habitats of special significance, Wildlife habitat, Rare, threatened or endangered species, Spawning, reproduction, and development, Marine habitat, Shellfish harvesting, and Estuarine habitat Excepted from Municipal and Domestic Supply

Requirements of this Order implement the Basin Plans **of this Regional Water Board and the San Diego Regional Water Board.**

40. Attachment F, Applicable Plans, Policies, and Regulations, page F-21, add Section III.C.10., as follows:

10. Water Quality Control Policy for the Enclosed Bays and Estuaries of California. This policy was amended by SWRCB's Resolution No. 95-84 on November 16, 1995 and prohibits the discharge of new discharges of municipal wastewaters and industrial process waters (exclusive of cooling water discharges) to enclosed bays and estuaries, other than the San Francisco Bay-Delta system, which are not consistently treated and discharged in a manner that would enhance the quality of receiving waters above that which would occur in the absence of the discharge. In order to comply with this policy, the Discharger must offset TN and TP loads (lbs/rainfall year) discharged under emergency conditions from the MWRP to Newport Bay at a minimum ratio 1.5:1 in accordance with an approved TN and TP offset plan. The Discharger has committed to minimize the TN and TP loads discharged to Reach 1 of San Diego Creek by implementing enhanced nutrient removal with operational objectives of producing recycled water with a monthly average total nitrogen and total phosphorus concentrations of 10 mg/L and 0.3 mg/L respectively during diversions. The MWRP will be operated in such a manner that the Discharger will meet these TN and TP effluent concentrations as an emergency discharge facility performance goal.

41. Attachment F, Applicable Plans, Policies, and Regulations, page F-22, replace the second paragraph of Nutrient TMDL of Section III.E., as follows:

Like prior Order No. R8-2007-0003, this Order implements relevant requirements of the nutrient TMDL. **The nutrient TMDL excludes total nitrogen (TN) loads from sources during storm events that result in flows of 50 mean daily cubic feet per second or more in San Diego Creek at Campus Drive, however, total phosphorous (TP) loads are not excluded under any flow conditions. To implement the nutrient TMDL, this Order includes a TN effluent limit of 1**

mg/L for stormwater, recycled water and tertiary treated effluent discharges to San Diego Creek, which is tributary to Newport Bay. TN and TP loads associated with the emergency discharge of stormwater or tertiary treated effluent (from MWRP) and recycled water overflows (from Sand Canyon Reservoir) to San Diego Creek, which cannot be excluded and are in excess of 1 mg/L for TN, must be offset at a minimum ratio of 1.5:1. This Order includes Discharge Prohibition III.F.4. that addresses this condition by requiring an approved nutrient offset program for recycled water overflows from Sand Canyon Reservoir to San Diego Creek in excess of 1 mg/L of TN. To implement the nutrient TMDL, this Order includes a total nitrogen effluent limit of 1 mg/L for stormwater discharges to San Diego Creek, which is tributary to Newport Bay. The Discharger has complied and proposes to continue to comply with this requirement by implementing a nitrogen offset program. Nitrogen discharges in excess of 1 mg/L will be offset by nitrogen reductions as the result of the Discharger's diversion and treatment of San Diego Creek flows in the IRWD constructed wetlands in the San Joaquin Freshwater Marsh. This Order includes Discharge Prohibition III.F.3, which addresses this condition by requiring an approved nutrient offset program for discharges exceeding 1 mg/L of total nitrogen. Also, during wet weather conditions, storm water runoff from the Sand Canyon Reservoir's watershed occasionally causes overflows³ from the reservoir into San Diego Creek. The Discharger has implemented best management practices to minimize overflows from Sand Canyon Reservoir during the rainy season. During the term of Order No. R8-2007-0003, Sand Canyon Reservoir has overflowed twice in 2010 and 2011 during the rainy season. The nutrient TMDL excludes nitrogen loads from sources during storm events that result in flows of 50 mean daily cubic feet per second or more in San Diego Creek at Campus Drive. Recent monitoring data indicate that the nutrient TMDL goals for 2012 have already been met. **The Discharger has complied and proposes to continue to comply with these requirements by implementing an approved TN and TP offset program as required by this Order.**

The TN and TP offset program will consist of computing and reporting the TN and TP loads that the Discharger removes from the San Diego Creek and Newport Bay Watershed, during the rainfall year (October through September), through the operations of IRWD's natural treatment system (NTS) wetlands, including the San Joaquin Freshwater Marsh and urban runoff diversions. TN and TP mass loads discharged to San Diego Creek must be offset with TN and TP mass removal credits accumulated within a three-rainfall year period that includes the rainfall year in which the discharge occurred. IRWD's treatment wetlands in the San Joaquin Marsh have removed nearly one million pounds of nitrogen since beginning full operation in 1999. More recent annual nitrogen removals have decrease due to declining flow rates and nitrogen concentrations in San Diego Creek, but the wetlands still removed 28,700 lbs of nitrogen in 2016-17. IRWD has documented up to 50,000 lbs of additional nitrogen removals for 2016-17 from its other NTS facilities in the San Diego Creek watershed. Unlike nitrogen, the IRWD wetlands in the San Joaquin Marsh do not provide

reliable phosphorus removal. However, IRWD's other NTS facilities removed a total of 1,181 lbs of phosphorus in 2016-17 and has removed a cumulative total of 18,914 lbs of phosphorus since beginning full operation in 1999. The 1.5:1 offset ratio is required to comply with the prohibition on new treated wastewater discharges included in the Water Quality Control Policy for the Enclosed Bays and Estuaries of California, which requires that new treated wastewater discharges enhance the water quality of the receiving water body (Newport Bay).

42. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-25, replace the title of Section IV.C., as follows:

C. Water Quality-Based Effluent Limitations (WQBELs) for DPs 003, 004, & 007 & Storm-007 (Tertiary treated effluent)

43. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-26 and F-27, replace the paragraphs for Total Chlorine Residual, TDS/Ammonia-Nitrogen and Table 12 of Section IV.C.2.a., as follows:

Total Chlorine Residual (TCR): Recycled water from the MWRP is distributed to the Sand Canyon, Syphon, and Rattlesnake Reservoirs for storage. Super-chlorination of the recycled water is necessary to prevent fouling of the recycled water distribution lines. As such, discharges of recycled water into the reservoirs contain concentrations of residual chlorine that may be toxic to aquatic organisms that may be present in the reservoirs. However, the residual chlorine present in the discharges dissipates within the reservoirs such that, currently, the reservoirs support aquatic life, including fish. The discharge of chlorinated recycled water to the reservoirs does not compromise the beneficial uses of the reservoirs, which include warm water aquatic habitat. The discharge of chlorinated recycled water to the reservoirs is necessary to accommodate wastewater reclamation and water conservation. It is in the public interest to accommodate these activities.

This Order specifies an effluent limitation for TCR to protect aquatic life beneficial uses. Compliance with this limitation, at Rattlesnake and Sand Canyon Reservoirs, is to be achieved outside a zone of initial dilution (about 45 feet away from the outfall at each reservoir), wherein dissipation of the chlorine occurs. In the case of Syphon Reservoir, due its smaller volume with respect to the other two reservoirs, compliance is to be achieved at the point of discharge to the reservoir (DP 007). **For tertiary treated effluent discharges through Storm-007, compliance with the TCR effluent limit shall be achieved at the discharge point to San Diego Creek.**

TDS/Ammonia-Nitrogen: TDS/Ammonia-Nitrogen limitations are specified in the Order for discharges to surface waters, including Sand Canyon, Syphon, and Rattlesnake Reservoirs **and Reach 1 of San Diego Creek**. The proposed TDS limit for the reservoirs is based on the prior effluent limit in Order No. R8-2007-

0003 and for emergency tertiary treated effluent discharges to Reach 1 of San Diego Creek is 1,500 mg/L. The proposed total ammonia-nitrogen monthly average limit was updated and calculated using the equations included in Table 4-4 of Chapter 4 of the Basin Plan. The numeric value of the proposed total ammonia-nitrogen limit is less than the limit included in Order No. 2007-0003 and details regarding the calculation of the proposed limit are included in Attachment O of this Order. For short-duration emergency discharge of tertiary treated effluent to Reach 1 of San Diego Creek the chronic criteria derived effluent limit applies to protect the WARM beneficial use of the reservoirs and as a facility performance criterion. The TDS/ Total Ammonia-Nitrogen limits are shown in the table below.

Table 12. TDS/Ammonia-Nitrogen Limits

Discharge Point	TDS, mg/L	Total Ammonia-Nitrogen, mg/L
003, 004 & 007	720	0.75
Storm-007	1,500	0.75

44. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-30, replace title of Table 14., as follows:

Table 14. RPA Summary for DPs 003, 004, & 007 & Storm-007

45. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-31, replace title of Section IV.D., as follows:

D. Best Professional Judgment-Based Effluent Specifications for DPs 003, 004, & 007 & Storm-007

46. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-32, replace title of Section IV.E., as follows:

E. Summary of Final Effluent Limitations for DPs 003, 004, & 007 & Storm-007

47. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-32, replace Section IV.E.2., as follows:

2. Satisfaction of Antidegradation Policy

Discharges in conformance with the requirements of this Order will not result in a lowering of water quality and therefore conform to antidegradation requirements specified in Resolution No. 68-16, which incorporates the federal antidegradation policy at 40 CFR 131.12. IRWD has significantly improved the quality of domestic water supply to its service area through construction and operation of the Irvine Desalter Project (see section II.A.5.), reduced reliance on high TDS imported water and use of low TDS

groundwater. During dry weather, all the treated effluent from these Facilities is and will continue to be recycled or in the case of LAWRP, the treated effluent will be discharged to the Pacific Ocean. During periods of low demand for recycled water at MWRP, wastewater that is not needed for recycling is diverted to OCSD facilities for treatment and ocean disposal or to the Orange County Water District's Groundwater Replenishment System. Implementation of the Groundwater Replenishment Project, a joint effort of the OCSD (which includes the Discharger) and the Orange County Water District has created and will create additional demand for recycled water. **The Discharger will comply with the prohibition for new treated wastewater discharges included in the Water Quality Control Policy for the Enclosed Bays and Estuaries of California by offsetting TN and TP loads (lbs/rainfall year) discharged under temporary emergency conditions from the MWRP to Reach 1 of San Diego Creek, which is tributary to Newport Bay, at a minimum ratio of 1.5:1 in accordance with an approved TN and TP offset plan. The Discharger has committed to minimize the TN and TP loads that could potentially be discharged to Newport Bay by implementing enhanced nutrient removal with operational objectives of producing recycled water with a monthly average total nitrogen and total phosphorus concentrations of 10 mg/L and 0.3 mg/L respectively during diversions. The MWRP will be operated in such a manner that the Discharger will meet these TN and TP effluent concentrations as an emergency discharge facility performance goal. The Water Quality Control Policy for the Enclosed Bays and Estuaries of California requires that new treated wastewater discharges enhance the water quality of the receiving water body (Newport Bay) if authorized.** The limitations in this Order are established at levels that assure no degradation of groundwater will occur as the result of wastewater recycling. Therefore, this discharge is consistent with federal and state antidegradation policies.

48. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-33, replace title of Section IV.E.4. and Table 16., as follows:

4. Summary of Final Effluent Limitations for DPs 003, 004, & 007 & Storm-007:

Table 16. Summary of Water Quality-based Final Effluent Limitations for Discharge Points 003, 004, & 007 & Storm-007

Parameter	Units	Effluent Limitations					Basis
		Average Monthly or as noted herein	Average Weekly	Max Daily	Instantaneous Minimum	Instantaneous Maximum	
BOD ₅	mg/L	20	30	--	--	--	BPJ,PO
Total Suspended	mg/L	20	30	--	--	--	BPJ,PO

Table 16. Summary of Water Quality-based Final Effluent Limitations for Discharge Points 003, 004, & 007 & Storm-007

Parameter	Units	Effluent Limitations					Basis
		Average Monthly or as noted herein	Average Weekly	Max Daily	Instantaneous Minimum	Instantaneous Maximum	
Solids							
Total Residual Chlorine	mg/L	--	--	--	--	0.1	PO, BP
Ammonia Nitrogen	mg/L	0.75					BP
TDS	mg/L	720/1,500 (12-M avg)	--	--	--	--	PO, BP
pH	Std. unit	--	--	--	6.5	8.5	PO, BP
Coliform	MPN	--	2.2 Median of last 7 days	--	--	--	PO, Title 22
Dichlorobromo methane	µg/L	46		71			CTR, SIP

Notes: PO = Previous Order, R8-2007-0003; BP= Basin Plan. **The 1,500 mg/L TDS limit applies only at Storm-007.**

49. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-34, replace Section IV.H.3. and add Table 17, as follows:

3. **Summary of Effluent Limitations at DPs 001, 002 and 008:** The TDS limitation of 910 mg/L for recycled water reuse on sites overlying the Irvine Groundwater Management Zone is based on the TDS water quality objective in Table 5-3 of the Basin Plan. **Also, the TDS limitation of 1,000 mg/L for recycled water reuse on sites overlying the Aliso Creek Hydrologic Sub-area are based on the TDS water quality objective included in the Basin Plan of the San Diego Regional Water Board. The effluent limits for biochemical oxygen demand and total suspended solids are based on best professional judgment (Section IV.D of Attachment F of this Order) and pH is based on water quality objectives included in the Basin Plans of this Regional Water Board and the San Diego Regional Water Board. All other recycled water effluent limits included in Table 17 below are based on the water quality objectives that apply to the Aliso Creek Hydrologic Sub-area; which are included in the Basin Plan of the San Diego Regional Water Board.**

Table 17 Recycled Water Effluent Limitations at DPs 001, 002 & 008

Parameter	Units	Effluent Limitations
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		Average Monthly	Average Weekly	12-month running average
Biochemical Oxygen Demand 5-day @ 20°C	mg/L	20	30	--
Total Suspended Solids	mg/L	20	30	--
pH	standard units	6 (instantaneous minimum)	9 (instantaneous maximum)	--
Chloride	mg/L			375
Sulfate	mg/L			375
% Sodium	%			60
Nitrate	mg/L			45
Iron	mg/L			0.3
Manganese	mg/L			0.05
MBAS	mg/L			0.5
Boron	mg/L			0.75
Aluminum	mg/L			1.0
Arsenic	mg/L			0.05
Antimony	mg/L			0.006
Barium	mg/L			1.0
Beryllium	mg/L			0.004
Cadmium	mg/L			0.005
Chromium	mg/L			0.05
Cyanide	mg/L			0.2
Mercury	mg/L			0.002
Nickel	mg/L			0.1
Selenium	mg/L			0.05
Thallium	mg/L			0.002

Note: The 12-month running average effluent limitations on this table apply only to recycled water use that overlies the Aliso Creek Hydrologic Sub-area within the San Diego Regional Water Board jurisdictional boundary.

50. Attachment F, Rationale for Effluent Limitations and Discharge Specifications, page F-35, renumber Table 17 as Table 18, as follows:

Table 1817. Stormwater Discharge Limits at Storm-007 & Storm-008

51. Attachment F, Rationale for Receiving Water Limitations, page F-36, replace Section V.A.2., as follows:

2. The previous Order R8-2007-0003 allowed discharges from Sand Canyon Reservoir under certain storm conditions. This Order also allows discharges under extreme storm conditions. The nutrient TMDL excludes nitrogen loads from sources during storm events that result in flows of 50 mean daily cubic feet per second or more in San Diego Creek at Campus Drive. This Order includes Discharge Prohibition III.F.3-III.F.4, which addresses this condition by requiring an approved nutrient offset program for **recycled water** discharges exceeding 1 mg/L of total nitrogen. **In addition, to address the potential of a temporary emergency discharge of tertiary treated effluent from MWRP to Reach 1 of San Diego Creek, which is tributary to Newport Bay, the**

Order requires the Discharger to implement an approved TN and TP offset plan that can mitigate TN and TP loads discharged to Newport Bay at a ratio of 1.5:1 and would minimize the TN and TP loads discharged by operating the MWRP in accordance with a facility performance goal. This Order specifies that this type of episodic emergency discharge is authorized under emergency conditions resulting from extreme rainfall events and/or season, as last resort disposal and after the Discharger has exhausted its recycled water storage capacity, has maximized the disposal of recycled water through OCSD's ocean outfall and it is not feasible to dispose of sewage through OCSD's sewage treatment plants (amongst other conditions). This Order recognizes the efforts of the Discharger to expand its recycled water storage capacity and work to evaluate the possibility of a permanent emergency discharge point to Reach 1 of the Santa Ana River, which would not have the current regulatory constraints.

52. Attachment F, Rationale for Receiving Water Limitations, page F-36, replace Section V.B., as follows:

C. Groundwater

The receiving groundwater limitations in the proposed Order are based upon the water quality objectives contained in the Basin Plans of this Regional Water Board and the San Diego Regional Water Board.

53. Attachment F, Rationale for Provisions, page F-40, add Section VII.B.2.d., as follows:

d. Discharger is required to submit for approval by the Executive Officer a TN and TP Offset plan that details the manner in which all pertinent information will be gathered to document the removal credits of TN and TP loads in the NTS, including the San Joaquin Freshwater Marsh and urban runoff diversions, managed by IRWD and the reporting of the data (See Section VI.C.2.e. of this Order).

54. Attachment F, Rationale for Provisions, page F-41, add Section VII.B.6.a., as follows:

a. Emergency Conditions for Discharge of Tertiary Treated Effluent to San Diego Creek: The Discharger is authorized to discharge tertiary treated effluent to Reach 1 of San Diego Creek, through discharge point Storm-007, on a temporary emergency basis, provided that the Discharger meets several conditions (See Section VI.C.6.b. of this Order).

55. Attachments; add Attachment P - San Diego Regional Board's Designation Agreement Letter, pages P-1 through P-4, which is attached to this Order.
56. Attachments; add Attachment Q - Santa Ana Regional Board's Designation Agreement Letter, pages Q-1 through Q-3, which is attached to this Order.

RECOMMENDATION:

Adopt Order No. R8-2018-0070, as presented.

COMMENT SOLICITATION:

Comments were solicited from the discharger and the following agencies:

U.S. Environmental Protection Agency (WTR-5) – Elizabeth Sablad
U.S. Army District, Los Angeles, Corps of Engineers - Regulatory Branch
U.S. Fish and Wildlife Service, Carlsbad
State Water Resources Control Board, Division of Water Quality – Afrooz Farsimadan
State Water Resources Control Board, Office of the Chief Counsel – Teresita Sablan
State Water Resources Control Board, Division of Drinking Water – Oliver Pacifico
California Regional Water Quality Control Board, San Diego Region, Brandi Outwin-Beals
State Department of Fish and Wildlife - Los Alamitos
State Department of Water Resources - Glendale
South Orange County Wastewater Authority – Brennon Flahive
Orange County Health Care Agency, Larry Brennler
Orange County Water District – Jason Dadakis
Orange County Sanitation District – Jim Colston
El Toro Water District – Dennis Cafferty
Southern California Association of Governments – Hasan Ikhrata
The Irvine Company – Dean Kirk
Orange County, Public Works – Chris Crompton
City of Newport Beach – City Manager
City of Irvine – City Manager
Orange County Coastkeeper - Garry Brown
Lawyers for Clean Water – Daniel Cooper